A Representation-Aware Late Fusion Framework for Generalizable Atrial Fibrillation Detection From Single-Lead ECG

A Dissertation

Submitted in partial fulfillment of the
requirements for the award of the degree
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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the dissertation report entitled **"A Representation-Aware Late Fusion Framework for Generalizable Atrial Fibrillation Detection From Single-Lead ECG"** in partial fulfillment of the requirements for the award of the Degree of Master of Technology and submitted to the **Department of Computer Science and Engineering** of Indian Institute of Information Technology Bhopal is an authentic record of my own work carried out during the period from **August, 2024** to **July, 2026** under the supervision of **Dr. Neha Singh** (Assistant Professor, Department of ECE), **Dr. Rahul Kumar Gangwar** (Assistant Professor, Department of Physics), and **Dr. Gajendra Babu** (Assistant Professor, Department of Mathematics) at Indian Institute of Information Technology Bhopal, India.

The matter presented in this dissertation has not been submitted by me for the award of any other degree of this or any other Institute.

Piyush S Wandile

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. Neha Singh	**Dr. Rahul Kumar Gangwar**	**Dr. Gajendra Babu**
Supervisor	Co-Supervisor	Co-Supervisor

The Disseration Viva – Voce Examination of **Mr. Piyush S Wandile** (24P02F1008) has been held on **July 31, 2026**.

Signature of External Examiner **HoD / Department Coordinator**

ABSTRACT

Cardiovascular diseases (CVDs) constitute a burgeoning public health crisis worldwide and continue to be among the most prevalent causes of morbidity and mortality irrespective of race and ethnicity. Prevention of such a crisis necessitates real-time, accurate, and prolonged monitoring of cardiac electrical signals in order to detect any abnormality prior to the occurrence of the disease. Although prolonged continuous monitoring used to be difficult, single-lead ECGs that have been incorporated seamlessly within Internet of Medical Things (IoMT) wearable devices have become a very promising approach. But processing of continuous single-lead ECGs into meaningful clinical alerts has been computationally challenging. This is especially the case when it comes to atrial fibrillation (AF), which is a clinically serious arrhythmia that poses a high risk of stroke and heart failure with electrophysiological characteristics of instability and multimodality.

Historically, the automatic detection systems used temporal representation of the ECG signal employing 1D CNNs for extraction of the morphological features. Though computationally efficient, such a system was proved to be unable to accurately represent the highly dynamic nature of the spectral properties of fibrillary waves, exhibiting a rather poor performance during its application to clinical tasks characterized by great patient-device variability. The primary hypothesis in this study consists of the assumption that introduction of heterogeneous feature spaces will significantly increase the robustness of the model to the changes in the domain and to the effect of noise. For that purpose, the proposed architecture will analyze the data from three different domains: the Time domain (based on 1D ECG signals for RR-interval analysis); Time-frequency domain (Short-Time Fourier Transform for local spectral density analysis); Time-scale domain (Continuous Wavelet Transforms for multi-resolution morphology analysis).

The initial model design, tuning of hyperparameters, and internal cross-validation was carried out using the PhysioNet/CinC 2017 challenge data set. To provide thorough validation of the framework's clinical applicability and generalization ability, an external cross-validation was performed using the totally invisible MIT-BIH AF data set, and inference was done without applying any secondary optimization or transfer learning. The results of our experiments clearly show that it is hard for deep learning models, which are trained on a single index representation, to generalize from one data set to another and thus have significant drops in both recall and specificity. On the other hand, the use of the suggested delayed fusion approach allows us to dynamically weigh probability estimates of the basis network for each of the domains and significantly outperform single-representation techniques. The presented quantitative approach provides us with 89.25% of accuracy and geometric mean 89.10% on the data containing invisible outliers and content-specific noises.

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Piyush S Wandile
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List of Abbreviations

1D CNN	:	One-Dimensional Convolutional Neural Network
2D CNN	:	Two-Dimensional Convolutional Neural Network
AF	:	Atrial Fibrillation
AFDB	:	Atrial Fibrillation Database
AUC	:	Area Under the Curve
AV	:	Atrioventricular
BiGRU	:	Bidirectional Gated Recurrent Unit
BiLSTM	:	Bidirectional Long Short-Term Memory
CinC	:	Computing in Cardiology
CNN	:	Convolutional Neural Network
CVD	:	Cardiovascular Disease
CWT	:	Continuous Wavelet Transform
DL	:	Deep Learning
ECG	:	Electrocardiogram
EMG	:	Electromyographic
FFT	:	Fast Fourier Transform
FN	:	False Negative
FP	:	False Positive
GAN	:	Generative Adversarial Network
G-Mean	:	Geometric Mean
GRU	:	Gated Recurrent Unit
HRV	:	Heart Rate Variability
IoMT	:	Internet of Medical Things
IoT	:	Internet of Things

LSTM	:	Long Short-Term Memory
LTAFDB	:	Long-Term Atrial Fibrillation Database
MLP	:	Multilayer Perceptron
NSR	:	Normal Sinus Rhythm
PAC	:	Premature Atrial Contraction
PR	:	Precision-Recall
PVC	:	Premature Ventricular Contraction
ResNet	:	Residual Network
RNN	:	Recurrent Neural Network
ROC	:	Receiver Operating Characteristic
RR	:	R–R Interval
SA	:	Sinoatrial
SMOTE	:	Synthetic Minority Oversampling Technique
STFT	:	Short-Time Fourier Transform
TN	:	True Negative
TP	:	True Positive
VF	:	Ventricular Fibrillation
VT	:	Ventricular Tachycardia

Chapter 1

Introduction

1.1 Background and Motivation

Cardiovascular diseases (CVDs) are universally recognized as a major cause of morbidity and mortality worldwide; World Health Organization estimates that this factor accounts for about 35% of annual deaths worldwide [1]. Placing a considerable economic and operational burden on international healthcare systems [2]. Traditionally associated with population aging, CVDs have undergone a highly concerning paradigm shift in recent decades. A marked and statistically significant increase in the incidence of sudden cardiac events, such as sudden cardiac arrest, myocardial infarction, and early arrhythmias, in much younger and previously asymptomatic populations has been observed [3]. A proactive stance is required regarding the treatment of heart disease because of the changing demographics. The primary reason for conducting this study is because of the meeting point between the pressing requirement for early detection and the inadequacy of technology currently being used. Continuous longitudinal monitoring is essential for detecting sporadic paroxysmal arrhythmias, which often occur outside the clinical setting and are missed during standard hospital assessments [4]. This continuous monitoring naturally generates a substantial and uninterrupted flow of physiological data. Manual interpretation of such amounts of data on a beat-by-beat level is practically impossible for cardiologists. Although applying DL algorithms to automate this analysis and remove the noise to detect clinically meaningful events is highly promising, however success is not assured. The feasibility of adopting an automated approach for portable or resource-constrained cardiac monitors remains to be determined.

1.2 Clinical Importance of Automated ECG Analysis

The ECG is a fundamental and widely used non-invasive diagnostic tool for recording and monitoring the complex electrical activity of the heart over time. Conventional diagnosis in emergency situations relies almost entirely on the 12-lead ECG, which provides a comprehensive, high-resolution, three-dimensional spatial view of cardiac electrophysiology. However, its operation requires precise electrode placement by a trained technician, making it computationally intensive, burdensome for the patient, and entirely unsuitable for continuous and prolonged ambulatory monitoring [5]. Single-lead systems are significantly less complex, much less expensive to manufacture, and ideally suited for integration into wearable devices (such as smartwatches and dedicated chest straps). Considering the practical advantages, the smooth incorporation of single-lead ECGs in the current Internet of Medical Things (IoMT) wearable technology has become one of the most promising ways of continuous patient monitoring [6]. Atrial fibrillation (AF), which is one of the most prevalent arrhythmias, with high chances of leading to strokes and heart failures, is highly unstable, multimodal, and shows unstable electrophysiology [7]. It is characterized by chaotic electrical discharges, rendering blood pumping ineffective [8]. Furthermore, it contributes significantly and independently to serious medical complications, including ischemic stroke and progressive heart failure [8]. Since atrial fibrillation can be asymptomatic and intermittent, automated systems capable of continuously monitoring data from a single ECG to accurately identify atrial fibrillation episodes are essential to issue early warnings and thus preventing fatal consequences [9].

1.3 Research Gap: Limitations of Single-Representation Learning

Even with the fast development of DL approaches in biomedical signal processing applications, their practical implementation is still a big issue. The main reason is that the electrophysiological processes in AF patients are characterized by the great instability and polyphony of the dynamics of the disease. Critical examination of current scientific literature allows us to detect some important problems of automated approach.

1.3.1 The limitations of temporal analysis

However, the existing research mostly applies 1D CNNs, which are designed to accept the raw unprocessed waveform data of the ECG, despite the fact that the 1D structure is considered an elegant and computationally light architecture. Temporal representations still struggle to capture the inherent spectral complexity and multi-frequency characteristics of filamentary waves [10]. Literature pertaining to foundational knowledge regarding biomedical signals states that although raw temporal signals are easy to compute, they do not effectively separate complex physiological oscillations from baseline wander and high frequency signals [11]. Time-domain models excel at identifying distinct morphological features such as R peaks, but they often fail when these features are masked by high-frequency noise or when the signal is dominated by chaotic, low-amplitude f-waves. These models exhibit remarkably poor generalization across different datasets, failing due to their inability to adapt to inter-patient morphological variation, different hardware acquisition noise profiles. For a complete understanding of the drawback, an analysis of individual studies involving the specific 1D architectures utilized in recent literature has been carried out in Chapter 3 of this thesis to clearly identify why temporal-only models fail when applied clinically.

1.3.2 The fragility of the single-domain transformations

To overcome the limitations of the time domain and directly address signal instability issues, many researchers have opted to convert 1D ECG time series into two-dimensional time-frequency representations before model training. Using transformations such as STFT spectrograms or CWT scalograms, researchers utilize 2D CNNs and time-frequency/scale domains to simultaneously exploit spatial correlations [12] [13]. The conversion from one-dimensional signals to their two-dimensional representation through the use of wavelets provides essential localization in frequency for non-stationary signals; however, using only one basis function tends to mask signals that are not mathematically compatible with the selected wavelet family [14]. While these 2D representations significantly improve the model's ability to distinguish within a single, isolated dataset, a glaring limitation remains that models trained exclusively on a single type of mathematical representation are inherently fragile. A careful assessment of each author's methodology as stated in Chapter 3 is the key to showing how use of particular 2D transformations in isolation is vulnerable to domain shifts caused by hardware.

1.3.3 Inadequate dataset validation

Many studies train and evaluate their models on exact subsets of the same dataset (e.g., using only the MIT-BIH or PhysioNet databases). However, ECG morphology varies considerably depending on patient demographics, acquisition equipment, ambient noise levels, and different medical annotation styles [15]. Models trained and tested on the same data distribution often experience a catastrophic degradation in performance when exposed to new data [16] [17]. This is a ubiquitous phenomenon known to us, such as domain switching in training sets [18]. Studies reporting near-perfect accuracy have seen their overall F1-scores plummet when evaluated on inconsistent external datasets [19]. This underscores that a model limited to a single signal representation is rarely robust enough to operate reliably in dynamic, real-world clinical situations with diverse patient populations.

1.4 Objectives of the Dissertation

The main objective of this research is to conceptualize, systematically design, and rigorously validate a generalizable DL framework capable of reliably identifying transient episodes of AF in diverse and novel datasets.

1. **Systematic evaluation and comparison of multi-domain feature representations:** Conducting a rigorous comparative analysis of DL models run in different computing spaces, specifically evaluating the effectiveness of the time domain (raw 1D ECG), the time-frequency domain (i.e. STFT here), and the time-scale domain (i.e. CWT here) within a strictly unified experimental framework.

2. **Development of a novel representation-aware late fusion architecture:** Designing an intelligent DL late fusion model that dynamically integrates, processes, and weights complementary diagnostic information from multiple distinct signal domains, thereby mitigating the weaknesses inherent in any single representation.

3. **Validation of generalizability and robustness with respect to external data sets:** Validation of the suggested multivariate approach with respect to an external validation process utilizing independent reference data sets. It entails evaluating the performance of the model on external, conflicting data sets acquired from other domains to prove its ability to withstand changes to the domain.

1.5 Major Contributions of This Work

This thesis makes great contributions to the field of automated arrhythmia detection by going beyond the limitations of the single representation learner model. The key contributions of the thesis are listed below.

- **Comprehensive Evaluation of Multiple Representations:** This study provides a comprehensive empirical analysis of the discriminatory performance of a specific DL architecture, applied to specific physiological data formats. It directly compares the feature extraction capabilities of RNNs on raw ECGs with those of spatial vision models (ResNet50, U-Net) in STFT and CWT representations.

- **Methodological Critique of Existing Approaches:** Offering a comprehensive and author-driven critique of current state-of-the-art deep learning pipelines in order to clearly identify the inherent weaknesses of single domain models, which become the immediate foundation of late fusion methodology.

- **Design of a Delayed Fusion Framework:** This research presents the design, implementation, and optimization of a representation-sensitive, specific delayed fusion architecture. A lightweight MLP is used to dynamically weigh probabilistic inferences generated from heterogeneous baseline models.

- **Empirical evidence of cross-dataset generalization and robustness:** Through rigorous external cross-validation, this study provides empirical evidence that the proposed fusion model offers a highly stable and balanced sensitivity and specificity profile. The results show that the integrated approach is significantly more resilient towards the domain changes induced by hardware characteristics than current state-of-the-art single-representation models.

1.6 Scope of the Work

The defined scope of this research deliberately aims to ensure methodological rigor and direct relevance to modern ambulatory care technologies. The study includes the design, implementation, and rigorous evaluation of cross-datasets for an automated AF detection pipeline. To ensure strict architectural compatibility with wearable IoMT devices, the research focuses specifically on the analysis of single-lead ECG signals. The use of a standard 12-lead medical ECG is obviously outside the scope of this work. Furthermore, model development, the initial training phase, and internal hyperparameter validation are strictly limited to the PhysioNet/CinC Challenge 2017 dataset. To preserve the integrity of the generalization test, the MIT-BIH AF dataset is completely excluded from the training phase and used only for external cross-validation. Furthermore, even though the system is theoretically flexible, at present it is restricted only to the classification of Normal Sinus Rhythm vs . Multi-class recognition of different kinds of verticular arrhythmia cannot be covered at the moment but may be trained along the same line of thought.

1.7 Organization of the Dissertation

The remainder of this thesis is carefully divided into multiple chapters in order to ensure a clear and logical flow of the research.

Chapter 2 (ECG Fundamentals and AF): This chapter provides the physiological rationale for the study. In this chapter, the electrical conducting system of the heart is explained. The morphology of the P-QRS-T waves seen in an ECG is described. More importantly, the pathophysiology of AF is discussed.

Chapter 3 (Literature Review): In this chapter, a comprehensive critical analysis of current approaches towards ECG analysis automation is given. In place of a thematic overview, the progress in this area is reviewed via an in-depth examination of existing approaches based on portable, 1D and 2D DL methods, organized by author. This fine-grained structure of the review was purposefully adopted to emphasize the specific limitations of the methodology and validation procedures in particular works, leading up to a comprehensive table at the end.

Chapter 4 (Datasets Used and Data Preparation): In this chapter, there is a complete statistical and structural analysis of the data sets utilized in this study. There is the description of the attributes associated with the standard PhysioNet/CinC 2017 data set employed in the study and MIT-BIH data set used for external validation.

Chapter 5 (Methodology and Framework Design): This chapter constitutes the technical heart of the thesis. This chapter provides the details regarding the suggested methodology starting from the signal preprocessing steps and the mathematical transformation techniques to create the multi-domain representations, STFT, and CWT, all the way to the analyzed DL architectures and the particular design principles of the delayed fusion methodology.

Chapter 6 (Results and Comparative Analysis): The analysis in this chapter provides a quantitative examination of the experiments performed. This includes the analysis of the performance measures of each reference model, and those achieved by the late fusion model. It concludes with a direct comparison with the leading-edge paper on a cross-dataset basis.

Chapter 7 (Conclusion and Future Scope): The concluding chapter provides a summary of the study, a conclusive response to the research objectives set out in the thesis, and highlights the shortcomings of the computation model suggested. It also suggests possible ways forward and future directions for research.

Chapter 2

ECG Fundamentals and Atrial Fibrillation

2.1 Electrical Activity of the Human Heart

The human heart is a complex biomechanical muscle pump that operates as an efficient biomechanical contraction due to an intricate internal electrical conduction mechanism that controls its operation. This electrical activity, which operates continuously and independently, forms the basic principle behind rhythmic circulation of blood in the body [20].

The electrical signaling within the cell occurs due to the presence of action potential. The fast changes of the electrical potentials across the cell membrane that are controlled by the flow of ions like Na+, K+ and Ca2+ through the membranes are the cause for this phenomenon. The normal sinus rhythm, medically called NSR, is believed to be generated by the SA node [21].

1. **Atrial Depolarization:** Often considered the primary natural pacemaker, the SA node is a specialized group of cells located in the upper wall of the right atrium. The highly coordinated electrical sequence that governs the generation and propagation of these action potentials through the heart muscle (myocardium) and that determines the progressive contraction (systole) and relaxation (diastole) of the heart chambers-forms a very orderly and predictable loop. It propagates uniformly through the right and left atria. This coordinated depolarization wave causes the simultaneous contraction of the atrial muscle tissue, actively propelling the remaining volume of blood towards the corresponding ventricle [21].

2. **Delay at the Atrioventricular (AV) node after atrial contraction:** The electrical impulse focuses on the AV node, which is positioned at the key point where the atrium and the ventricle meet. What is essential to remember about the electrical

signal at this point is that there is a considerable physiological delay experienced by the nerve impulse (0.1 seconds). Such a delay is relevant mathematically and physiologically due to the fact that it enables the ventricles to be filled with the blood (end-diastolic volume) before the electrical stimulation of contraction [20] [21].

3. **Ventricular Depolarization:** It takes place when there is delay in the conduction of AV node, and nerve impulses conduct very fast through the bundle of His, splitting into the right and left bundle branch and finally into the Purkinje fibres. Rapid conduction of the nerve impulses through the thick network causes very powerful, massive and coordinated contraction of the myocardium of the ventricles which pushes the deoxygenated blood towards the lungs through the right ventricle and oxygenated blood to the systemic circulation through the left ventricle [20].

4. **Repolarization:** The ventricle must then actively pump ions across its membranes to return to its initial position. This process, called repolarization, prepares the cardiac tissue for the next action potential and the subsequent cardiac cycle [21].

2.1.1 Cellular Electrophysiology and the Action Potential

In order to understand the chaotic electrical activity observed in AF, it is essential to examine the cardiac action potential at the cellular level. Myocardial contraction depends on the transmembrane movement of key ions, including Na^+ (sodium), K^+ (potassium), and Ca^{2+} (calcium) [21].

As can be seen from Fig. 2.1, there are five stages that make up the cardiac action potential. **Phase 0** is associated with depolarization and involves a rapid increase in Na^+ ion concentration that results in membrane potential elevation from $-90\,\text{mV}$ to $+20\,\text{mV}$.

Phase 1 corresponds to early repolarization, and there is a transient outward current of K^+ ions at this phase. Five phases make up a typical cardiac action potential. **Phase 0** corresponds to fast depolarization, during which there is an influx of Na^+ ions causing a transmembrane potential change from $-90\,\text{mV}$ to about $+20\,\text{mV}$. **Phase 1** is the first phase of repolarization that occurs due to a transient outward K^+ current.

Phase 2 is referred to as the plateau phase, and it is specifically unique to the heart. It features an interaction between the inward movement of Ca^{2+} and outward movement of K^+ [20]. **Phase 3** represents the rapid repolarization phase, characterized by the movement of potassium ions out of the cell.

Automaticity occurs due to slow and spontaneous depolarization that takes place during **Phase 4** in SA node cells. However, in the presence of diseases such as AF, remodeling takes place in the structure and dysfunction of channels, particularly the delayed rectifier potassium channels, leading to reduced action potential and effective refractory periods [21]. The prolonged episodes of arrhythmias lead to extensive cell remodeling causing changes in the structure and function of ion channels in the myocardium, leading to the worsening of the irregular propagation of the action potentials [22].

This allows for the propagation of multiple wavelets simultaneously in the atria. This causes production of f-waves in the ECG [22].

Table 2.1: Common Arrhythmias and Their Defining ECG Characteristics

Arrhythmia Type	Heart Rate / Rhythm	Key ECG Characteristics
Bradycardia	$<$ 60 bpm, regular or slightly irregular	Normal P waves and QRS complexes; prolonged RR intervals; slow but organized rhythm
Tachycardia	$>$ 100 bpm, usually regular	Rapid rate; P waves may merge with preceding T waves; shortened RR intervals
Atrial Fibrillation (AFib)	Irregularly irregular rhythm	Absence of distinct P waves; presence of fibrillatory (f) waves; variable RR intervals
Atrial Flutter	Atrial rate $\sim$ 250–350 bpm	Sawtooth flutter waves (F waves); regular atrial activity; variable AV conduction
Ventricular Tachycardia (VT)	$>$ 120 bpm, regular	Wide QRS complexes; absent or dissociated P waves; monomorphic or polymorphic patterns
Ventricular Fibrillation (VF)	Chaotic, no measurable rate	No identifiable P, QRS, or T waves; disorganized electrical activity; irregular waveform
Premature Ventricular Contractions (PVCs)	Underlying rhythm with premature beats	Early, wide, and bizarre QRS complexes; no preceding P wave; compensatory pause
Premature Atrial Contractions (PACs)	Irregular rhythm due to early beats	Early P wave with abnormal morphology; normal QRS; incomplete compensatory pause

To contextualize AF within the broader spectrum of cardiac rhythm disturbances, Table 2.1 presents a summary of common arrhythmias along with their characteristic ECG features.

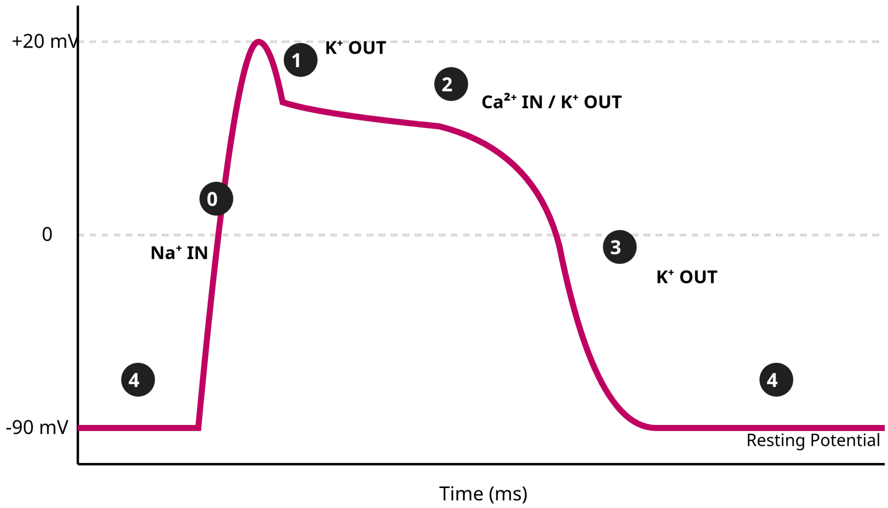


Figure 2.1: Figure showing the normal cardiac action potential, depicting the five different stages (0-4) of cell depolarization and repolarization along with their respective ionic currents through the membrane.

2.2 Single-Lead ECG Signal Acquisition and Characteristics

ECG is a standard non-invasive medical diagnostic technique used to continuously record heart electrical activity across the skin. While conventional clinical diagnosis relies heavily on the full 12-lead ECG, a high-resolution, three-dimensional tracing using 10 physical electrodes for complete visualization, this technique has practical limitations. The 12-lead configuration is highly sensitive to patient movement, requires trained clinical staff for implementation, and is complex and computationally expensive for continuous, long-term ambulatory monitoring [5] [6].

With the advent of miniaturized wearable health technologies, the single-lead ECG has unquestionably become the standard for remote ambulatory monitoring. Single-lead configuration that is typically designed to replicate lead I of Einthoven's triangle to record the electrical potential differences between two distinct points on the body [11]. This method is significantly less complex, far less invasive, manufactured and administered via patches placed on the arm or chest, and considerably less expensive. The single-lead system is ideally suited for continuous, real-time monitoring of heart rhythm [23]. Image acquisition over several days or even weeks results in a significantly reduced spatial resolution, inherent to single-lead recordings.

This presents highly specific and complex diagnostic challenges. By limiting their analysis to a single electrical vector, these systems capture far less morphological information about structural abnormalities of the heart muscle. Most important of all, single-lead configurations are inherently more sensitive to environmental noises, respiratory baseline changes, high frequency motion artifacts and major intra-patient structural variability. Unlike Holter monitoring which is performed clinically, modern smartphones and wearable devices are prone to corruption due to external noise interference and body movements and need carefully developed feature extraction algorithms that can operate with single lead information [24] [25]. As a result, the use of highly robust and representative image signal processing algorithms becomes necessary for the effective extraction of these mathematical features from such simplistic and noisy signals. Since there are no standardized methods for obtaining these parameters in clinical applications, each individual approach needs to be examined separately to determine its influence on the resulting classification.

2.3 Standard ECG Waveform and Its Components (P-QRS-T)

The sequential electrical events of the cardiac cycle are graphically represented on the surface of ECG as a distinct sequence of aberrations (waves) and specific isoelectric periods (intervals). A typical, healthy cardiac cycle is classically represented by the P-QRS-T complex, as depicted in Fig. 2.2 [21]:

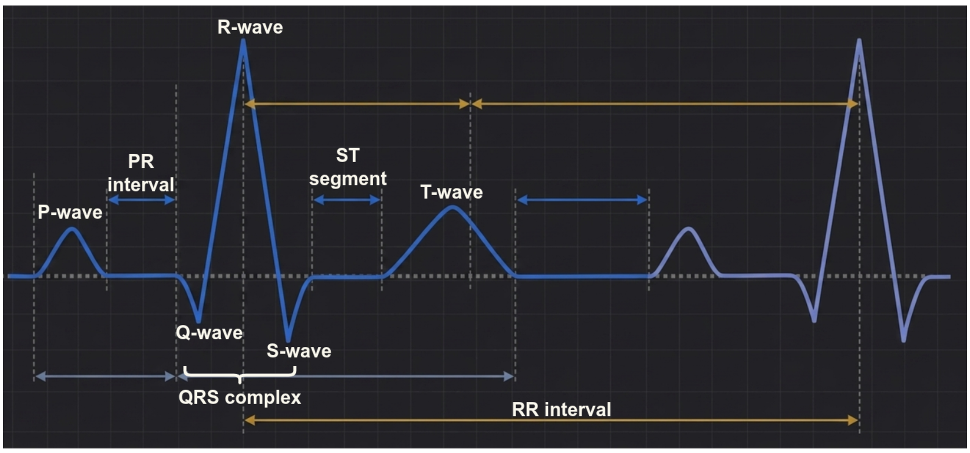


Figure 2.2: Morphology of a healthy person's ECG wave, showing the P-QRS-T wave pattern, normal physiological amplitudes, and isoelectric intervals.

- **P wave:** P wave is coherent, with a regular and smooth contour, representing depolarization, indicating that the electrical impulse normally originates from the SA node, reflecting a NSR.

- **PR interval:** This is a crucial interval measured precisely between the onset of the P wave and the QRS complex, which is reflected at the AV node.
- **QRS complex:** This is the most prominent, high-frequency, high-amplitude feature of the ECG. It is accompanied by rapid and powerful depolarization of the right and left ventricles. Morphologically, it comprises a Q wave (early negative downward deviation), an R wave (strong and significant positive upward deviation), and an S wave (subsequent negative downward deviation) [11].
- **ST Segment:** This is the horizontal isoelectric line connecting the end of the QRS complex (J point) to the beginning of the T wave. This segment represents the initial plateau phase of ventricular repolarization, during which the ventricles are fully contracted but have not yet begun to re-establish their electrical activity.
- **T Wave:** These features of the T wave such as amplitude, shape, and polarity form important medical factors and may also be used as early detection signs for ischemia or electrolyte imbalance [20].

Small deviations in these parameters of the component waves help recognize abnormalities in heart functioning. It is important to conduct an analysis to identify and extract long-term dependencies between waves over several cardiac cycles.

2.4 Pathophysiology of AF

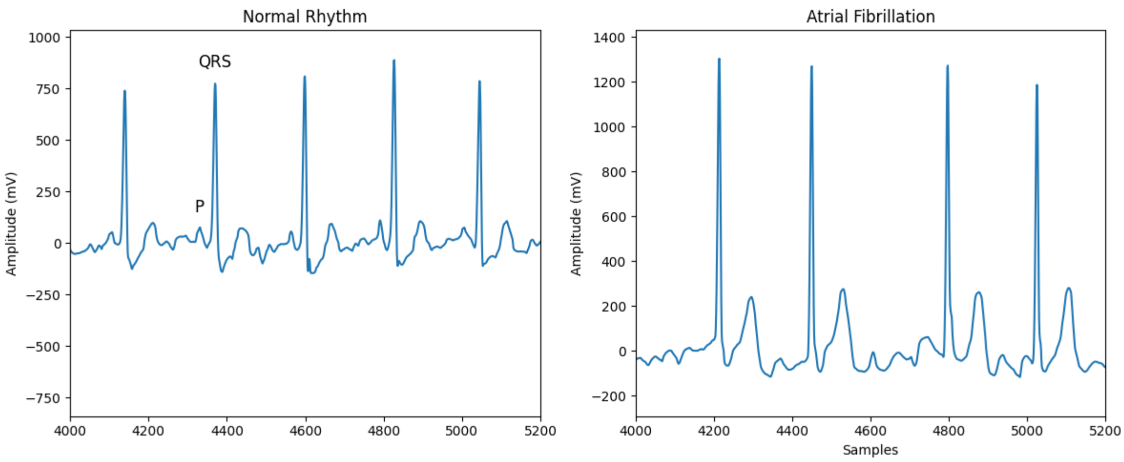
AF is known to be among the most prevalent cardiac arrhythmias that pose a great concern both clinically and globally, contributing significantly to the load in intensive care units. In terms of pathology, AF is considered a complete disturbance in the cardiac electric physiology. This disease is associated with highly abnormal and erratic electrical discharges in the atrial cells [8].

Unlike the synchronized depolarization process emanating from the SA node, which occurs at one point, AF results from multiple spontaneous and simultaneous firing of ectopic electrodes located close to the pulmonary veins [20]. This leads to the electrical disarray that makes efficient contraction for pumping blood impossible.

In this situation, the AV node, which is responsible for the fibrosis (involving fast, uncoordinated muscle contractions) of the atrial muscle tissue, is hit by hundreds of impulses each minute. These impulses are erratic in nature. However, as a safety measure, this AV node does not allow many of these impulses to go through but transmits some of them randomly [21]. The irregular transmission of impulses leads to an extremely erratic ventricular reaction.

In hemodynamics, lack of effectiveness in atrial contractility leads to pooling and stagnation of blood in the atria. In case this occurs persistently and remains untreated, it is bound to lead to development of other major diseases such as blood clots, which will then circulate through the body and ultimately cause stroke or heart attack [20]. In case left untreated, the disordered electrical impulses associated with atrial fibrillation may result in extreme hemodynamic instability, increasing risks of sudden cardiac arrest and death [3].

2.5 Non-Stationary Dynamics and Artifacts in AF Episodes



(a) Normal episode

(b) AF episode

Figure 2.3: ECG morphology comparison of single lead recording: (a) Normal Sinus Rhythm (NSR) showing prominent P waves with regular R-R interval, and (b) An episode of Atrial Fibrillation (AF) where there is no regular P wave seen, erratic fibrillatory f-waves and highly irregular R-R intervals are noted.

As visually contrasted in Fig. 2.3, the presence of AF is manifested by very distinct and mathematically quantifiable temporal variations in its morphology on a single-lead ECG. The most distinctive diagnostic features of AF are the typical regular P waves (often replaced by continuous low and high-frequency fibrous F waves) and, in extreme cases, the complete absence of irregular R-R intervals (the distance between successive ventricular contractions) [8].

Unlike NSR, which exhibits a high degree of mathematical periodicity, statistical stability, and a steady state, AF displays marked and highly unstable behavior. The intermittent nature of paroxysmal AF, in which the heart rapidly switches from normal rhythm to fibrillation within short, 30-second ECG segments, is designed to distinguish a normal ECG from the one affected by AF.

Combined with the subtlety of F-wave expression, this diagnostic complexity makes automated diagnosis particularly challenging. This complexity is compounded by the fact that single-lead dynamic ambulatory ECGs are inherently very noisy. These devices are highly sensitive to low-frequency baseline variations (due to patient respiration and changes in electrical impedance), high-frequency EMG noise from the contraction of underlying skeletal muscles, and large-amplitude transient motion artifacts [26]. Such devices which surround us everywhere have the potential to pick up F-waves or create artificial interference with artificially short R-R intervals. Thus, existing rule-based algorithms for detecting heartbeats are highly sensitive to errors [6].

2.6 Challenges faced in AF Detection via Short Lead ECGs

A few concrete difficulties arise that need to be resolved in order to successfully employ the use of automated DL techniques for detecting AF based on short single-lead ECG signals.

First of all, standard signal processing techniques that are generally dependent solely on R-peak detection rules (such as the popular Pan-Tompkins rule), are not usually effective when there are severe problems with the signal caused by motion artifacts and baseline drift [11].

Secondly, due to its complex polyphony and inherent instability, AF poses a challenge in terms of signal representation in the form of 1D temporal representations, where it becomes increasingly difficult to accurately represent the spectral complexity of the pathological activity [10]. Since the amplitude associated with the fibrotic activities is not very high, therefore, small spectral variations between two consecutive QRS complexes become invisible. In order to develop a generalizable model, it becomes computationally necessary that the signal processing must be done with respect to such signals which are defined by both temporal and spectral properties [27] [28]. Additionally, because of the fact that the nature of the fibrillation signal is non-stationary and arbitrary in its morphology, therefore, according to recent literature, it has been established that there should be such a representation model which incorporates various domains into one [16] [29]. Although the data fusion models may have some theoretical approaches, their implementation on practical grounds is quite custom. It is extremely important to understand the constraints that previous researches have faced in order to create an effective late fusion method.

2.7 Variance in Methodology and Evaluation Approach

With regards to the theoretical underpinnings of artificial intelligence for ECG signal analysis, there is no limit to the number of combinations that can be done in terms of preprocessing, feature extraction, and classification. The reason being, datasets pertaining to the cardiovascular system experience domain shift due to hardware problems. Thus, the performance of the model will depend upon these various micro-decisions by the authors in their respective works.

Chapter 3

Literature Survey

There have been many paradigm changes in recent years regarding automated identification of AF through single-lead ECGs. The present chapter will provide an in-depth study of various contributions towards signal processing, DL architectures, and detection of arrhythmias in the field of biomedical engineering. This analysis focuses on the specific architectural methods employed, the nature of the training data used, and the generalizability of the results obtained, in relation to the objectives of this thesis.

3.1 State-of-the-Art Techniques in Automated ECG Analysis

Historically, automatic heart rhythm classification relied almost entirely on explicit feature engineering algorithms. These early computation methods required domain experts to manually define the rules for calculating heart rate variability (HRV), measuring R-R interval irregularities, or mathematically identifying P waves. However, this traditional paradigm has evolved considerably and has ultimately been largely replaced by DL paradigms.

DL models eliminate the need for manual heuristics, instead of aiming to extract extraordinarily complex, nonlinear discriminating features directly from raw physiological data. While the integration of DL has led to undeniable improvements in classification accuracy on isolated and highly controlled datasets, the current situation more precisely illustrates the ongoing challenge of reconciling computational efficiency (necessary for performance planning) with robust and reliable generalization of data across different datasets (necessary for diverse clinical populations). In order to understand precisely why this problem of generalization exists, it becomes essential to break down the exact topological and preprocessing decisions taken by leading researchers in the coming sections.

3.2 Lightweight and Wearable-Oriented Approaches

A significant and growing body of recent research highlights the critical need for computational efficiency, with the explicit goal of deploying predictive models in resource-constrained environments, such as battery-powered wearable devices for the Internet of Things (IoT) and medical technology.

The work of Menaceur et al. in this area is remarkable. They proposed a highly specific hybrid DL architecture designed to reduce computational load utilizing DSC-BiGRU [17]. Their method combines depth-separable convolutions to reduce the number of parameters, stimulus compression attention mechanisms to focus on important temporal features, residual combinations to avoid gradient extinction, and biGRUs to capture sequential dependence. The pipeline places emphasis on simplicity, with use of a balance between SMOTE and layer-wise cross-validation using patients from the MIT-BIH datasets. While extremely successful, external studies have found these simplistic models to trade off high recall for precision when confronted with novel noise.

In the same manner, the work done by He et al. presents another compact ML pipeline tailored for portable devices [24]. In this work, a classifier, such as XGBoost, which is less computationally expensive than DNNs, is preferred. Feature extraction depends largely on wavelet denoising, algorithmic heartbeat detection, signal inversion correction, and mathematical grouping of the RR interval values as inputs to the classifier. This technique managed to achieve a good F1 score of about 0.85. Although these light-weight approaches seem quite appealing for embedded systems, there is a serious limitation with them; that is, the dependence of their performance on heavy preprocessing (such as QRS complex detection algorithms).

3.3 One-Dimensional DL Models (Time-Domain)

In order to overcome the weaknesses of explicit feature extraction and peak detection algorithms, the vast majority of current researches focus on employing 1D convolutional neural networks. Such an approach is based on directly utilizing raw 1D ECG data as inputs for such kinds of models. 1D neural networks are the dominant choice in many biomedical tasks owing to their simplicity and extremely low computational costs.

For instance, Phukan et al. presented a model known as AFibri-Net, which is a very specific convolutional neural network designed explicitly for running on resource-limited edge computing nodes [34]. In order to make sure that these models performed optimally, Serhani et al. used reinforcement learning algorithms to optimize the hyperparameters of the 1D CNN, obtaining excellent results on the MIT-BIH isolated arrhythmia dataset [35].

Empirical evidence has shown that 1D CNNs have been particularly adept in identifying localized patterns of temporality in physiological signals. Moreover, the effective incorporation of architectural regularizers like excessive application of dropout layers and weight decay methods have been found very efficient in improving the generalizability of such networks. However, as indicated by Ma et al., this method of time domain analysis has some major weaknesses, which have often attracted criticism. The major limitation associated with models based on purely temporal data involves the huge variation in morphology observed across different patients as well as the differences in acquisition noise present in different monitoring devices, which makes the model trained on raw time domain data ineffective in providing reliable results. In other words, time domain models have difficulties adapting to real-life inter-patient and inter-device variability. Moreover, as pointed out by Narotamo et al., although in rare cases, temporal models have been found superior to image-based models on some clean and noise free dataset, the former models perform poorly when dealing with complicated spectral variations and non-stationary automata [10]. Therefore, while being computationally simple, purely temporal data models fail to handle complexities of multifrequency cardiac events.

3.4 Time-Frequency and 2D Representation Approaches

To directly address the seemingly non-stationary nature of AF and overcome the spectral insensitivity of 1D models, many researchers have investigated modifications to complex 2D representations of the 1D ECG time-domain signal before integrating them into DL processing chains. The use of STFT spectrograms or CWT scalograms enables the use of powerful deep 2D CNNs often adapted from computer vision. These 2D networks are particularly effective at simultaneously exploiting spatial correlations in the time and frequency/amplitude domains.

The effectiveness of this method is well documented. Xie et al. reported substantial and statistically significant performance improvements when using CWT scalograms coupled with a highly complex multi-branch ResNet architecture, compared to their 1D counterparts evaluated on the same dataset [12]. Their study concluded that 1D models inherently fail to fully exploit the joint time-frequency information contained in the physiological signal (Mewar et al.). The use of 2D wavelet-encoded ECG signal images conclusively demonstrated that 2D CNNs handle the hyperstable, multitoned behavior of AF much more effectively than one-dimensional networks [13]. Most importantly, this 2D approach effectively avoids the need for tricky and error-prone preprocessing steps. Beyond algorithmic detection of R peaks, Anbalagan and Nath's research took this paradigm further by using a combination of STFT, Stockwell, and Chirplet signal representations, fed (via transfer learning) into large, pre-trained vision networks to capture very subtle spectral variations specific to AF [36].

3.5 Limitations of Existing Cross-Dataset Validation Studies

Despite the numerous theoretical and empirical advantages of using 2D transforms to capture spectral dynamics, the current literature presents a major limitation: profoundly inadequate cross-dataset validation. Most published studies use a single monolithic dataset (the PhysioNet database).

However, in the practical application of these models in clinical settings, the morphology of the ECG exhibits extreme unpredictability based on several factors, including patient characteristics, types of acquisition devices used, noise present in the environment, as well as the clinical notation system used by each organization. Domain switching refers to the fact that, despite training and testing DL models using the exact same statistical data distributions, their performance is catastrophically affected when applied to new data.

Such an example includes models trained on the Chapman data and tested on the MIT-BIH AF data set [17]. A complete breakdown of the models' effectiveness has been noted. While such models usually preserve their high accuracy (or even perfect), they have poor recall rates (for instance, 66.70%) leading to low overall F1-score results (80% or lower). The issue concerns not only 1D raw but 2D STFT data sets as well; neither type proves to be robust enough for practical usage.

3.6 Emerging Trends in DL Frameworks for Automatic ECG Analysis

Though ID and 2D CNN frameworks have been used mainly to perform automatic ECG analysis before, the era since 2023 till today has experienced a revolution towards the development of DL frameworks that can learn long-range dependencies and overcome the dependence on large annotated datasets.

3.6.1 Attention Mechanisms and Transformer Architectures

To address the inherent locality problems associated with the receptive fields of regular CNNs and vanishing gradients of deep RNNs, attention mechanisms and Transformer architectures have become increasingly popular in physiological signal processing. For instance, Park et al. effectively utilized self-attention to assign dynamic weights to certain temporal sections of ECGs in their arrhythmia classification approach to improve their assessment of diagnostic uncertainty [40]. Moreover, Menaceur et al. incorporated SE attention blocks in their lightweight BiGRU structure to enforce the model's ability to concentrate only on most informative physiological information and ignore the irrelevant background noise [17]. Going beyond simple sequence modeling, Jia et al. developed an advanced hybrid model composed of multi-scale CNN layers, bi-directional LSTMs, and full Transformers (MSB-ST) [31]. Self-attention from the Transformer turned out to be extremely effective at representing global spatio-temporal dynamics of the chaotic fibrillatory waves from the lengthy clinical records, although this is generally a computationally expensive approach.

3.6.2 Paradigms of Self-Supervised Learning (SSL)

The lack of high-quality data annotated by physicians is the biggest barrier to the success of DL models in medicine. SSL was invented as a cutting-edge method that can be applied to large amounts of unlabeled ECG data. Specifically, Zhu et al. proposed the development of a period-aware SSL framework which captures the dynamic relations of inter and intra-periods in cardiac cycle without any explicit clinician annotations in the pre-training stage [29]. The proposed approach greatly helped in feature extraction from unbalanced data sets. In addition, C. Ma et al. showed the clinical feasibility of SSL for wearable devices with the introduction of the SSL-based pretraining approach which extracts very robust mathematical representations from continuous noise-filled ambulatory data stream [26].

3.7 Justification for a Multimodal Late Fusion Approach

A comprehensive review of the literature on the most recent research reveals three major short-comings:

1. **Temporal Blindness:** One-dimensional models operating on raw ECGs are very efficient computationally, but they often miss the complex spectral properties of atrial fibers, and high frequency fibrillatory waves suffer from significant generalization problems when exposed to new data and noise profiles.

2. **Single-Domain Fragility:** Two-dimensional time-frequency (STFT) and close timescale (CWT) approaches greatly improve discrimination capabilities on closed datasets, but they still suffer from significant generalization problems when trained separately and with new noise and hardware profiles.

3. **Vulnerability to domain changes:** The robustness of cross-data is largely understudied and insufficiently addressed in the current literature, making most published models highly vulnerable to domain changes and practically unusable for IoMT deployment in real-world conditions.

This approach is justified. By systematically integrating complementary physiological information from polygenic fields, and by exploiting the temporal morphology of the raw signals, the high spectral density of STFT, and the local multiscale characteristics of CWT, the integrated and delayed fusion model is theoretically capable of significantly improving resilience to domain changes. This important gap remains to be filled by the systematic development, optimization, and external validation of the delayed fusion framework.

3.8 Comprehensive Summary of Related Works

The switch from classical digital signal processing techniques to DL networks has resulted in numerous network architectures for arrhythmia diagnosis. As can be seen in the above discussions, some of these methods range from simple one-dimensional temporal networks to complex two-dimensional vision networks based on spectral transformations. The following table 3.1 outlines a comprehensive timeline of the most influential studies in this field within the last ten years.

Table 3.1: Summary of State-of-the-Art DL Frameworks for AF Detection

Author & Year	Architecture	Signal Domain	Dataset Used	Key Metrics	Identified Limitations
Ng et al. [32]	pSi-ResNet (Siamese CNN)	1D (30s Segments)	MIT-BIH AFDB, Long-Term AFDB	F1: 96.84% – 97.07%	Requires patient-specific labels for few-shot learning; highly vulnerable to domain shift across general populations.

Continued on next page

Table 3.1 continued from previous page

Author & Year	Architecture	Signal Domain	Dataset Used	Key Metrics	Identified Limitations
Xie et al. [12]	CWT-MB-ResNet (Multi-Branch)	2D (CWT Scalograms)	OUHSC (PDF), PhysioNet 2017	F1: 0.886 (PhysioNet), 0.739 (OUHSC)	High computational complexity required for CWT transformations on longer clinical segments.
Rahul & Sharma [16]	Survey of AI Advancements	N/A	PhysioNet (MIT-BIH, PTB-XL)	N/A	Models show poor inter-subject generalization due to morphological variability; "black box" nature limits clinical trust.
Menaceur et al. [17]	DSC-BiGRU + SE Attention	1D (AV Conduction Ratios)	Chapman University	Acc: 98.23%, F1: 98.27%	Relies on a single database; heavily depends on visual validation (Grad-CAM) for interpretability.
Aashiq et al. [33]	Random Forest, GNB (TFDGiniXML)	2D (CW-TFD + Gini Index)	MIT-BIH Arrhythmia, Fantasia	Acc: 94.44%, F1: 95.24%	Relies on engineered Gini Index features rather than end-to-end DL feature extraction.
He et al. [24]	DBSCAN Clustering + XGBoost	1D (RR Intervals)	PhysioNet 2017, CPSC2021	F1: 0.847 (CinC), 0.843 (CPSC)	Heavy, restrictive reliance on explicit algorithmic clustering and manual feature extraction.

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Table 3.1 continued from previous page

Author & Year	Architecture	Signal Domain	Dataset Used	Key Metrics	Identified Limitations
Phukan et al. [34]	AFibri-Net (5-layer 1D-CNN)	1D (5s Segments)	6 Standard DBs, 3 Unseen DBs	Acc: 99.97%, Sens: 99.95%	Requires deeper energy consumption profiling before deployment across diverse IoT microcontrollers.
Serhani et al. [35]	RL-Optimized CNN	1D	MIT-BIH Arrhythmia	Acc: 97.40%, Loss: 0.0081	Reinforcement learning heuristic tuning introduces significant initial resource consumption overhead.
Mewada [13]	2D Haar DWT + CNN	2D (Wavelet Images)	MIT-BIH Arrhythmia	Acc: 99.52%, F1: 95.64%	Requires integration of fixed-weight wavelet filters directly into pooling layers to minimize parameters.
Anbalagan & Nath [36]	Ensembled DNNs (ShuffleNet+ AlexNet)	2D (STFT, Chirplet, Stockwell)	PAF Prediction, PhysioNet 2017	Acc: 93.70%, F1: 88.98%	Transforming 1D signals into 2D matrices adds significant computational overhead.
Lim & Jang (Kim et al.) [23]	SeqAFNet (Dual-RNN Many-to-Many)	1D (Beat-wise RR)	AFDB, LTAFDB, HUINNO Patch	High Consistency (IEC standards)	Wearable patch devices are highly susceptible to severe noise and signal vanishing.

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Table 3.1 continued from previous page

Author & Year	**Architecture**	**Signal Domain**	**Dataset Used**	**Key Metrics**	**Identified Limitations**
Liu(Ma et al.) [6]	Survey on Wearable AF Monitoring	1D	MIT-BIH, CPSC2021	N/A	PACs cause massive false positives; high dependency on QRS detection algorithms.
Narotamo et al. [10]	1D GRU vs. Multimodal Fusion	1D vs. 2D (GAF, MTF, RP)	PTB-XL (12-lead ECG)	G-Mean: 80.35% (1D), 79.42% (Fusion)	2D transforms and image resizing destroyed high-frequency features, adding overhead without performance benefits.
Yu et al. [30]	DDCNN (Dual-Channel VGG16)	1D (Morphology + HR)	PhysioNet 2017	Avg F1: 0.833, AF F1: 0.805	Extreme dataset imbalances necessitated highly aggressive synthetic noise injection and balancing.
Serhal et al. [27]	Survey (Wavelet vs. FFT methods)	Time-Frequency (Wavelet)	MIT-BIH Database	Acc: 99.23% (CNN), 100% (PNN)	Total volume of literature utilizing robust WT for AF prediction remains surprisingly low.

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Table 3.1 continued from previous page

Author & Year	Architecture	Signal Domain	Dataset Used	Key Metrics	Identified Limitations
Jia et al. [31]	MSB-ST (CNN + BiLSTM + Transformer)	1D (Local MS + Global ST)	AFDB, LTAFDB, NSRDB, SHDB	Acc: 98.98%, Sens: 99.48%	Single-lead design struggles with complex spatial coordination; requires expansion to 12-lead setups.
Park et al. [40]	Self-Attention LSTM-FCN	1D (Time Series)	PhysioNet 2017	Acc: 94.20%	High dependence on localized temporal ordering before applying attention mechanisms.
Menaceur et al. [17]	DSC-BiGRU + SE Attention	1D (AV Conduction Ratios)	Chapman University	Acc: 98.23%, F1: 98.27%	Evaluates using only one dataset; very susceptible to distributional changes.
Zhu et al. [29]	Period-Aware Self-Supervised Learning	1D (Periodic segments)	MIT-BIH, CPSC	F1: 91.20%	Depends entirely on period-aware temporal information without any spectral information.
Ma et al. [26]	Self-Supervised Pretraining	1D (Ambulatory ECG)	Wearable ECG DBs	Acc: 96.80%	Needs an extremely large amount of unlabeled physiological data for pretraining.

This discussion highlights the importance of a key insight derived from the above tabled data. In spite of the very good performance of isolated models in closed-loop datasets, they tend to have serious challenges when dealing with domain-shift problems. The main problem associated with 85% of the discussed research is that it fails to consider rigorous external validation between the test and training sets.

Chapter 4

Datasets Used and Data Preparation

The conceptualization, training, and empirical validation of a highly generalizable DL architecture require the use of diverse, detailed, and representative physiological datasets. A fundamental flaw in many contemporary biomedical machine learning studies is the over-reliance on a single homogeneous data source. To explicitly address and rigorously evaluate the proposed representation-sensitive delayed fusion framework, this research systematically uses two fundamentally different and globally recognized reference datasets: the PhysioNet CinC Challenge 2017 dataset, which serves as the primary engine for model development. Whereas the MIT-BIH AF dataset is strictly reserved for completely independent external cross-validation.

4.1 PhysioNet CinC Challenge 2017 Dataset (Primary Training/Validation)

The PhysioNet CinC Challenge 2017 dataset was specifically designed and organized by the international scientific community for automatic cardiac synchronization using single-lead ECG recordings, often small and noisy, acquired with consumer-grade wearable devices (data from the AliveCor handheld ECG device). This particular data set, which addresses today's pressing classification problems, is the basis of this study. This particular data set will be used as a single-source data set for training, validation, and benchmarking of each individual baseline model and the entire late fusion MLP neural network.

4.1.1 Dataset Characteristics and Signal Demographics

This data set is made up of a wide variety of thousands of individual ECG samples with rhythm annotations in a single lead. For maximum accuracy in terms of temporal resolution, all physiological data in this data set have been collected by sampling at a rate of 300 Hz. Individual samples vary significantly in duration, from very short 9 seconds long recordings up to recordings that last over 60 seconds. This variation in recording duration truly captures the unpredictability associated with recordings generated in real-world use conditions of wearable devices.

4.1.2 Strict Patient-Wise Splitting and Class Balancing Strategies

One of the most prevalent and critical errors made in current ECG classifiers is that of data leakage at the subject level. This issue is caused by the fact that the continuous signal of the same individual is split into overlapping windows that get accidentally distributed among the train and test sets. As a result, the machine learning model learns one baseline morphology of the person rather than general arrhythmias symptoms. A meticulous patient-level segmentation approach based on mathematically rigorous conditions was conducted. The data were systematically segregated into training, validation, and testing subsets with an exact ratio of 80:10:10.

Moreover, due to the fundamental statistical nature of physiological signals, the occurrence of regular beats is much higher than those of paroxysmal transitory AF. The dataset itself is inherently skewed due to its natural imbalances, despite the current machine learning literature attempting to artificially increase the minority classes. Although the new approach utilizes approaches for artificial data creation such as SMOTE [17] and GANs [37], the suggested technique specifically avoids such approaches [30].

The rationale for this deliberate omission of artificial data enrichment lies in the high physiological complexity of cardiac electrophysiology. Mathematical techniques such as SMOTE are fundamentally based on linear interpolation among the small number of instances present in the feature space. When this linear reasoning is blindly applied to nonlinear and highly complex physiological time-series data, or to their 2D spectral variations, the interpolation often produces morphologically impossible and clinically unintelligible waveforms. Similarly, while GANs excel at classic computer vision tasks, they struggle to accurately model chaotic, yet physically constrained, rhythms.

In practice, the synthetic ECG samples generated by these deep networks retain a biologically artificial appearance. It is virtually impossible for generative algorithms to faithfully reproduce the specific and real physiological randomness of a malformed human heart without introducing artifacts. If these artifacts were integrated into the training process, it will force the DL base models to associate their weights with computer-generated artificial noise. Which is bound to be quite different from the natural noise generated due to the human heart's electrical activity. To strictly preserve the absolute morphological and spectral integrity of the biological signals, the necessary class balance would be managed without altering the native data structure i.e., by using a strict weighting penalty for classes in the loss calculation or by directly and distortionlessly replicating a minority of cases. Used exclusively during the training phase, the validation and mathematically isolated testing subsets remained intact and unchanged in their natural state of disequilibrium, thus ensuring the absolute integrity and practical applicability of the final reported performance indicators.

4.2 MIT-BIH AF Dataset (External Cross-Validation)

One of the major weaknesses that often hampers innovation in current research in automated classification of ECGs – based on several systematic literature reviews – is the natural inability of deep neural networks to retain their accuracy and robustness in face of out-of-distribution data. Typically, training and testing of models are done using identical subsets of a homogenous database, including PhysioNet. As a result, neural network models that are solely trained and optimized using such single-source databases have been found to show poor performance in completely new settings. This problem arises from the fact that real-life settings are unique due to the presence of variable physical surroundings, distribution of ambient noises, variation in medical annotations, and distinct morphology for various patients. The problem is common during model transfer from one setting to another and has been referred to as domain shift; in such situations, models showing nearly perfect accuracy experience drastic drops in F1-scores.

In order to effectively address such a major weakness in an effective manner and validate the late fusion method's real-life generalizability using an empirical approach, the well-known MIT-BIH Atrial Fibrillation (AF) database was purposefully employed within the experimental setting. While the usual consumer-grade recording equipment used by patients to record their ECG activity usually results in very short recordings that range anywhere from 9 to 60 seconds, the highly reliable MIT-BIH database consists of continuous and long-term Holter monitor ECG records. This involves the recording of 10 hours of invaluable information on 25 human subjects with varying degrees of AF severity.

Additionally, one of the main differences between the hardware recording specifications in the above-mentioned external database and the hardware specifications of the initial training set is in the use of different acquisition specifications on purpose. Although the initial training data was acquired at 300 Hz, the physiological signals in the MIT-BIH database were acquired at a lower speed of 250 Hz. Moreover, the external database uses more precise 12-bit signal format, which can record the signals within the range of ±10 mV. Such purposeful choice of completely different hardware specifications for acquiring the data leads to an obvious domain shift.

It is therefore crucial to realize that this particular database has been strictly used as an external validation database throughout this whole process. The complete absence of any records in the primary training stage, the second stage of further tuning, and even hyperparameter optimization is crucial because it will guarantee the fact that the model has never seen this particular kind of noise pattern or patient structure before. Backpropagation of any kind and weight adjustments were not allowed in this particular case, which provides the highest level of authenticity and scientific integrity of the framework validation process.

4.2.1 Characteristics of the External Validation Data

Unlike the small, patient-initiated recordings in the PhysioNet dataset, the MIT-BIH AF database comprises extensive, continuous, long-term ambulatory Holter monitor ECG recordings from human subjects with varying degree and severity of AF. Parameters used in the native acquisition are completely different from those used in the primary training dataset since the physiological signals were acquired at a much slower sampling rate of 250 Hz with a 12-bit data format for an extremely wide dynamic range of ±10 mV. These recordings, derived from patient-specific demographics and vastly different clinical hardware environments compared to AliveCor's PhysioNet dataset, represent mathematically profound and real "domain shifts." Using a frozen and fully trained fusion model to make predictions on this dataset allows it to be recycled, refined, or adjusted after each training transfer. Without this, the proposed computing framework provides an exceptionally authentic and objective assessment of its performance during real-world clinical deployment across various diagnostic scenarios. A detailed parametric comparison highlighting this profound domain shift between the primary training and external validation datasets is provided in Table 4.1.

Table 4.1: Detailed Comparison of MIT-BIH AF and PhysioNet/CinC 2017 Datasets

Feature	**MIT-BIH AF Database**	**PhysioNet/CinC 2017**
Origin/Device	Ambulatory ECG recorders (Beth Israel Hospital)	AliveCor handheld device
Leads/Channels	Two-channel	Single-lead
Subject Selection	25 human subjects with AF (predominantly paroxysmal)	Unknown number of subjects; data donated by AliveCor users
Duration	10 hours per record	9–60 seconds
Dataset Size	25 records	8,528 (train) + 3,658 (test)
Sampling Rate	250 Hz	300 Hz
Resolution	12-bit (range: $\pm$10 mV)	Not specified
Target Classes	AFIB (AF), AFL (atrial flutter), J (AV junctional rhythm), N (normal/other)	Normal, AF, Other, Noisy
Annotation Strategy	Rhythm-level labeling	Global labeling

Chapter 5

Methodology and Framework Design

5.1 Overall System Architecture

The intelligent arrhythmia detection system is carefully designed and implemented as an end-to-end processing pipeline for computing. It is specially tailored to algorithmically process and handle raw and inherently volatile physiological signals recorded in the form of a single channel signal. This is because these dynamics cannot be handled using a conventional system due to the volatility involved. To counter the drawbacks posed by a single dimensional representation, the mathematical transformation of these standardized single dimensional signals produces a set of multi-dimensional multimodal representations. These state-of-the-art mathematical projections consistently model the rhythmical abnormalities in time, the fluctuations in spectral density in high frequencies through time-frequency analysis, and the morphological features at multiple resolutions in the scale-time space. These distinct feature spaces are first evaluated and classified separately using highly customized deep base networks, which include optimized RNNs for temporal sequence data, deep ResNet50 for spectral data, and U-net encoders for multi-resolution features in hierarchical space. In the post-isolated probabilistic inference stage, the framework then combines the predicted probability vectors produced from the evaluation process. This intelligent aggregation of the diagnosis is done using a representation aware late fusion scheme where a light weight MLP is used to compute the appropriate weights for the different modalities before providing a definite diagnosis. As shown in the detailed flowchart depicted in Fig. 5.1, the overall process flow is sequentially broken down into four major stages including rigorous pre-processing of all signals (including fixed time windows and Z-score normalization), domain transformation of the signal (time, STFT and CWT domain), individual training and optimization of DL models and intelligent MLP-based fusion of probabilities.

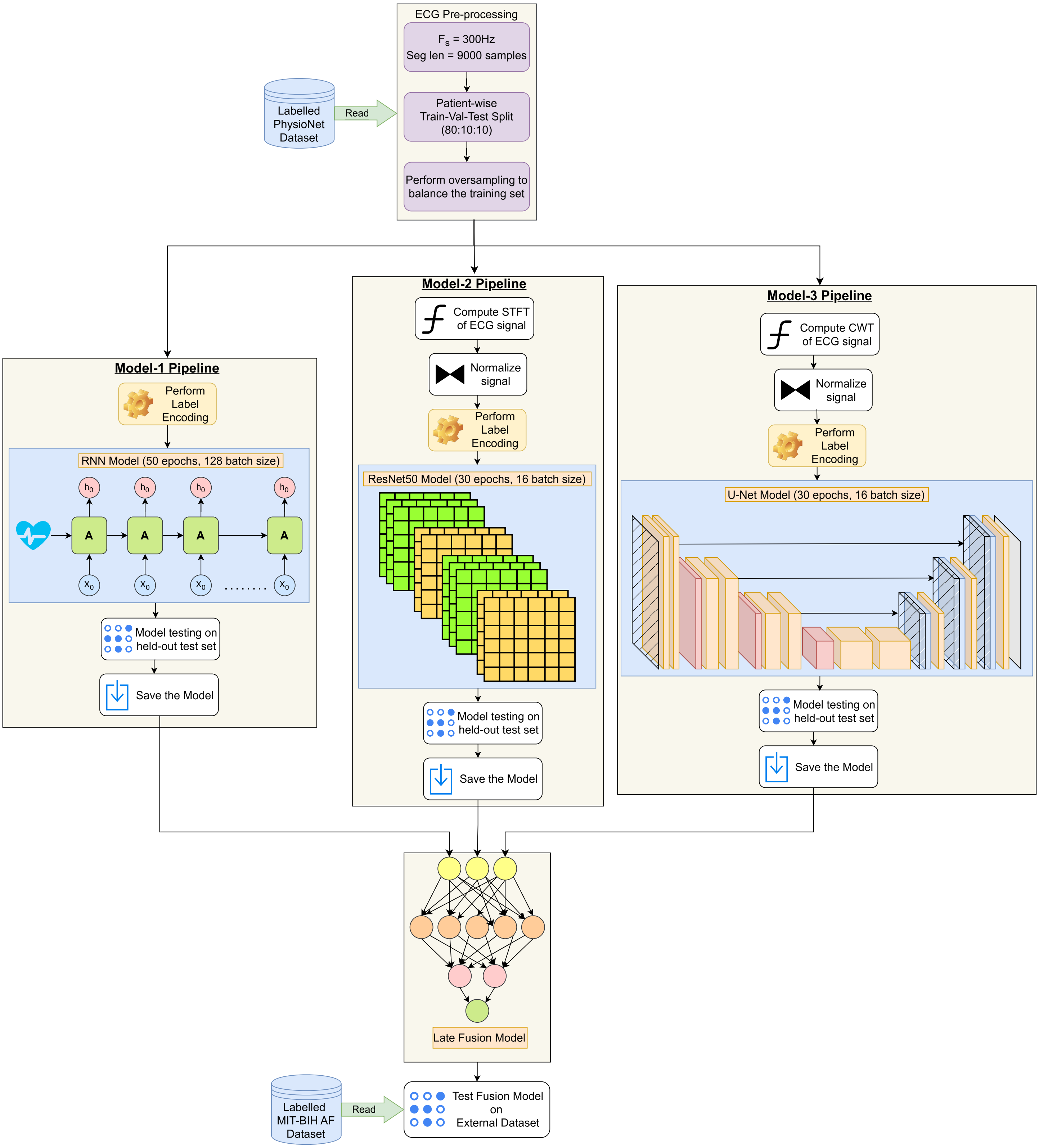


Figure 5.1: Proposed research methodology end-to-end workflow showing data pre-processing, three parallel neural networks pipeline (RNN, ResNet50, and U-Net) for feature extraction in specific domains, and late fusion approach tested on an external dataset.

5.2 ECG Signal Preprocessing and Windowing Strategy

Given the use of multiple heterogeneous datasets characterized by different native hardware acquisition parameters, the implementation of a universally consistent preprocessing pipeline is essential to ensure perfect mathematical consistency between all evaluated models, especially during the highly critical external evaluation phase across the entire dataset. The incoming ECG signals were strictly standardized in the unified, common input tensor format.

- **Algorithmic Resampling:** The core training data (PhysioNet) was natively acquired at a sampling rate of 300 Hz. To ensure accurate temporal dimension compatibility, the continuous recordings of the external validation data (MIT-BIH AF) were resampled from their original frequency of 250 Hz to 300 Hz using a mathematically robust interpolation algorithm.

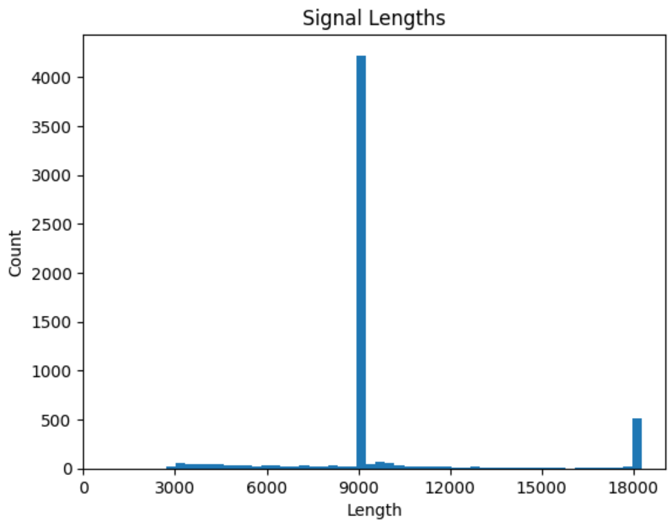


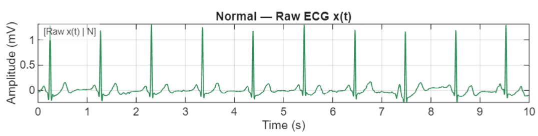
Figure 5.2: Length distribution of the ECG signal segments in the PhysioNet data set, showing that the length of 9,000 samples is the dominant one used for preprocessing.

- **Fixed-length windowing strategy:** As evidenced by the signal histogram shown in Fig. 5.2, the vast majority of signals cluster tightly around a specific length. Therefore, after thorough statistical analysis of the frequency distribution of continuous signal lengths present in the Core PhysioNet dataset, a mathematically rigorous fixed-length windowing strategy was adopted. Each continuous ECG recording was systematically divided into precisely controlled windows of 9,000 discrete samples, seemingly shorter or longer than the strict 30-second physiological recording time limit at 300 Hz. The recordings were rejected (if shorter) and truncated/split (if longer), ensuring a universally uniform one-dimensional input size of $9{,}000 \times 1$.

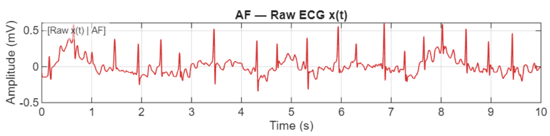
- **Static Z-score normalization:** The Z-score normalization data for DNNs, based on a normalized data distribution to avoid gradient peaks and accelerate convergence (arithmetic mean and standard deviation), were calculated using only the PhysioNet training data matrix. To prevent data leakage, these specific statistical parameters were frozen and logged. They were then universally applied as static transformations during the validation, internal testing, and external testing phases of the records from the MIT-BIH dataset.

5.3 Multimodal ECG Representations

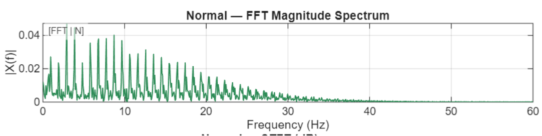
For computational modeling of the complicated nature of temporal dynamics of atrial fibers, which is highly non-stationary and constantly changing, the 9000×1 ECG signal vector was considered, based on PhysioNet dataset records, Normal: A01151, AF: A08242. The vector was processed in the time domain as well as mathematically transformed to three completely different and highly complementary feature spaces: FFT, STFT, and CWT., visually represented in Fig. 5.3.



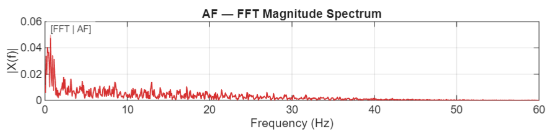
(a) Normal — Raw ECG



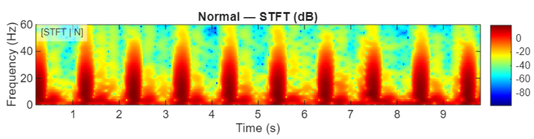
(b) AF — Raw ECG



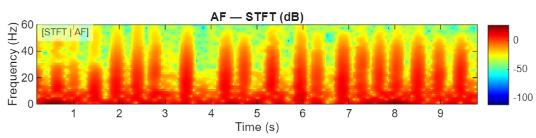
(c) Normal — FFT Magnitude Spectrum



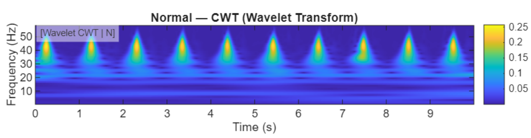
(d) AF — FFT Magnitude Spectrum



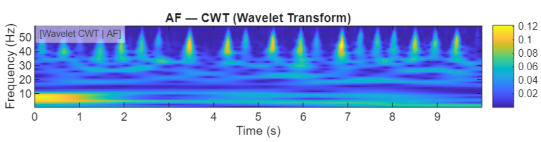
(e) Normal — STFT



(f) AF — STFT



(g) Normal — CWT



(h) AF — CWT

Figure 5.3: NSR (left) and AF (right) multi-domain comparison. In order to highlight morphological and spectral differences, the ECG waveforms are shown in four different domains: a, b – raw-time domain; c, d – FFT magnitude spectra; e, f – STFT spectrograms; g, h – CWT scalograms.

5.3.1 Raw ECG (Time-Domain)

From a computational perspective, the most direct and fundamental representation of a physiological signal is its raw, one-dimensional time-domain waveform. For basic time-domain models, the input tensor is a statically normalized numerical array of size 9000×1. The main advantage of this raw representation is that it perfectly preserves the original time-domain morphology of the P-QRS-T complex. By retaining this raw state, DL models can discover and map sequential rhythmic irregularities (such as highly variable R-R intervals) and ultimately learn without resorting to complex and artificial algorithmic feature extraction.

5.3.2 Short-Time Fourier Transform (Time-Frequency Domain)

Although the time domain is excellent for morphological mapping, it suffers significantly from spectral blindness. Mathematically, the STFT has been used to reveal and quantify the hidden high-frequency spectral variations of AF which are completely obscured when observed solely as a function of time. The STFT is particularly well suited to the analysis of volatile physiological signals. It involves mathematically dividing the continuous time signal into discrete, highly overlapping time windows and then calculating the FFT for each independent window. This transform is rigorously defined mathematically as:

$$X(p,\omega) = \sum_{q=-\infty}^{\infty} x[q]\, w[q-p]\, e^{-j\omega q} \tag{5.1}$$

Where *x[q]* represents the one-dimensional discrete time signal, *w[q-p]* denotes the sliding mathematical analysis window (which eliminates spectral leakage), and ω represents the angular frequency. The result is a 2D spectrogram - a dense visual matrix that accurately represents the evolution of the signal's complex frequency content over time. Common hyperparameters utilized in the generation of these spectrograms were a window size of 256, a hop size of 32 samples (in order to ensure high continuity in time) and zero-padding set to on. The output from this computation resulted in a rigid two-dimensional tensor shape of 129×283×1.

5.3.3 Continuous Wavelet Transform (Scale-Time Domain)

Although the STFT is powerful, it is inherently limited by the Gabor limit: it uses a fixed window size, thus imposing a permanent mathematical trade-off between time and frequency resolution. To overcome this limitation, the CWT was introduced. The CWT dynamically optimizes its operational resolution: it offers exceptionally high time resolution at high frequencies (ideal for separating sharp QRS complexes) and perfectly accurate frequency resolution at low frequencies (ideal for mapping basic f-waves). It is obtained by mathematically correlating the original signal with continuously scaled and translationally shifted versions of a specific mother wave.

The Complex Morlet wavelet for a specific biomedical application is precisely defined as a complex sinusoid exponentially modulated by a Gaussian envelope of a Morlet wavelet, chosen for its mathematically optimal joint localization in both time and frequency domains.

$$\psi(t) = \pi^{-1/4} e^{j\omega_0 t} e^{-t^2/2} \tag{5.2}$$

A super criterion of 64 distinct parameters was used to compute highly detailed timescales. To reduce the computational overhead required for deep 2D vision networks. The time axis of the resulting matrix is algorithmically down sampled by a factor of 2. The resulting final CWT image tensor is fixed at a highly dense shape of $64 \times 4500 \times 1$.

5.4 DL Model Architectures Evaluated

In order to ensure complete scientific rigour and eliminate any form of bias in methodology, several architectural models for DNNs have been exhaustively tested. Such a comprehensive test regime was intended to conduct systematic and extremely objective comparisons among three distinct signal representations, with the goal of discovering the best possible mathematical representation space for arrhythmia detection. The process of testing each architectural model involved iterated tests on the model, where both tests with and without the use of mathematical regularisation techniques were conducted; this included high dropout and L2 weight decay regularisation to prevent network overfitting. Finally, in order to assure reliability, 5-fold cross validation was carried out throughout all tests.

5.4.1 1D-Architectures (CNN, RNN, LSTM, BiLSTM, GRU, BiGRU)

In order to work on the raw time vectors without any losses of morphological information, specially trained 1D sequential models were used. One-Dimensional (1D) Convolutional Neural Networks (CNNs) were used as the default feature extractors since they have shown their capability to find and extract morphological features in a localized spatiotemporal manner, including certain peak abnormalities. Moreover, RNNs, along with access control based recurrent networks, namely LSTM and GRUs were comprehensively analyzed. Such RNN architectures can help us find complex non-linear and long-range temporal dependencies that span over tens of cardiac cycles and help in identifying prolonged arrhythmia. The exact architecture configuration with corresponding layers, convolution strides, and tensor shapes used for our selected 1D temporal RNN baseline are given in Table 5.1.

5.4.2 2D-Architectures (AlexNet, VGG16, ResNet18, ResNet50, U-Net)

To perform an effective analysis on the mathematically altered images, a range of highly parametrized computer vision networks was applied to tackle the dense and complicated 2D STFT spectrograms as well as CWT scalograms. The selected networks are classical examples of deep mesh networks like AlexNet, extremely deep networks like VGG16, as well as highly improved residual networks like ResNet18 and ResNet50 that rely on skip connections to deepen the training process. All of the selected computer vision models are particularly adept at exploring highly sophisticated spatial dependencies not only across the longitudinal time dimension but also along the vertical dimension of frequencies or amplitudes. Additionally, the analysis involved a unique implementation of an encoder in the form of a U-net that provided a particularly strong hierarchical feature extraction ability, especially valuable for analyzing mathematically altered images.

Table 5.1: Architectural Parameters of the Proposed 1D-RNN Model (Raw ECG)

Layer	**Kernel**	**Stride**	**Output Shape**	**Params**
Input	–	–	$(9000, 1)$	0
Conv1D (32)	7	2	$(4500, 32)$	256
BatchNorm	–	–	$(4500, 32)$	128
MaxPool	2	2	$(2250, 32)$	0
Conv1D (64)	5	2	$(1125, 64)$	10304
BatchNorm	–	–	$(1125, 64)$	256
MaxPool	2	2	$(562, 64)$	0
SimpleRNN (128)	–	–	$(562, 128)$	24704
BatchNorm	–	–	$(562, 128)$	512
Dropout	–	–	$(562, 128)$	0
SimpleRNN (64)	–	–	(64)	12352
BatchNorm	–	–	(64)	256
Dropout	–	–	(64)	0
Dense (128)	–	–	(128)	8320
BatchNorm	–	–	(128)	512
Dropout	–	–	(128)	0
Dense (1)	–	–	(1)	129

5.4.3 Architectural Deep-Dive: The ResNet50 Spectral Extractor

For the effective analysis of the complicated $129 \times 283 \times 1$ STFT spectrogram tensor, the Deep ResNet50 model with 50 layers was used as a special neural network architecture for extracting the spectral features. The increase in the number of layers in the networks leads to the appearance of various difficulties associated with the performance of classical CNNs. The most common among these is the vanishing gradient problem, which occurs in the course of backpropagation. In such a case, the error gradients become close to zero, and the training process is hampered.

As a result of making use of identity skip connections in an efficient manner, the ResNet model is thus able to effectively address and solve this optimization problem. With such a structure, both input and gradient signals can be directly transmitted over specific middle layers without passing through all the layers to ensure continuous mathematical computation. The mathematical structure of a residual unit is as follows:

$$w = \mathscr{F}(v, \{W_i\}) + v \tag{5.3}$$

In this critical equation, both v and w directly signify the input and output vectors of the provided block, while $\mathscr{F}$ is defined as the particular residual mapping function that the layers of the neural network need to learn mathematically. An extensive structural analysis illustrating the detailed architectural configurations, the exact strides applied for each convolution, and the final output dimension of the corresponding tensors is provided in Table 5.2 below.

Table 5.2: Architectural Parameters of the ResNet50 Model (STFT-Based)

Layer	Kernel	Stride	Output Shape	Params
Input	–	–	$(129, 283, 1)$	0
Conv2D (16)	3×3	1	$(129, 283, 16)$	160
BatchNorm	–	–	$(129, 283, 16)$	64
ResNet50 Backbone	–	–	$(4, 9, 2048)$	23,587,712
Flatten	–	–	(73728)	0
Dense (512)	–	–	(512)	37,749,248
Dropout	–	–	(512)	0
Dense (2)	–	–	(2)	1,026

5.4.4 Architectural Deep-Dive: The U-Net Scale-Time Encoder

In the case of CWT scalograms, the contraction path (encoder) of the U-Net architecture was utilized. U-Net is a fully convolutional network that applies two successive 3×3 convolutional layers, each followed by a rectified linear unit (ReLU) activation, and a 2×2 max pooling operation with stride 2 for downsampling at each stage.

In the case of every succeeding down-sampling operation within the encoder model architecture, the spatial dimensions of the feature maps get halved, whereas their depth in terms of the number of channels gets doubled intentionally. This particular model architecture, which involves a combination of spatial down-sampling operations along with simultaneous channel expansion, plays an extremely important role by ensuring that the model efficiently captures the highly complex and multi-resolutional features that exist in CWT scalograms. The use of aggressive channel expansion ensures that low-level textural features are converted into high-level spectral representations. The architectural configuration in this regard has been presented in Table 5.3.

Table 5.3: Architectural Parameters of the CWT-Based CNN Encoder

Layer	**Kernel**	**Stride**	**Output Shape**	**Params**
Input	–	–	$(64, 4500, 1)$	0
Conv2D (32)	3×3	1	$(64, 4500, 32)$	320
BatchNorm	–	–	$(64, 4500, 32)$	128
MaxPool	$(2, 4)$	$(2, 4)$	$(32, 1125, 32)$	0
Conv2D (64)	3×3	1	$(32, 1125, 64)$	18,496
BatchNorm	–	–	$(32, 1125, 64)$	256
MaxPool	$(2, 4)$	$(2, 4)$	$(16, 281, 64)$	0
Conv2D (8)	1×1	1	$(16, 281, 8)$	520
MaxPool	$(2, 4)$	$(2, 4)$	$(8, 70, 8)$	0
Flatten	–	–	(4480)	0
Dense (256)	–	–	(256)	1,147,136
Dense (2)	–	–	(2)	514

5.5 Model Selection Strategy

Mathematical efficiency and diagnostic reliability of the best neural network architecture for each physiological communication channel can be considered to be the primary computational aim of this thorough evaluation step. It is vital to state that the process of such a thorough selection is mainly governed by complex multivariate conditions based on the usage of specific mathematical measures of effectiveness, such as the F1 score, to establish an adequate balance between sensitivity and specificity amid the existing problems with class imbalance in clinical datasets. With the consideration of the highest level of discrimination and cross-subject stability as the major criteria for evaluation, the selected optimal models will be used as the key components for extracting features in the form of probabilities. These highly specialized neural networks will then function as the primary processing units within the proposed delayed fusion framework.

- **The F1-Score:** The necessity of strict multivariate selection criteria, mathematically defined as the main metric, was key to obtaining the optimal tradeoff between precision and recall.

- In addition, there is a need for maintaining a proper tradeoff between sensitivity and specificity owing to the imbalanced nature of physiological data.

- **Generalization Stability:** This is another important aspect regarding the stability of the performance metrics for each isolated and patient-specific testing set. Thus, based on this, a classical RNN design was selected for the ECG time domain, a ResNet50 architecture for the STFT time-frequency domain, and a U-net encoder for the CWT time-scale domain.

5.6 Proposed Representation-Aware Late Fusion Framework

The key computational feature behind this approach is the introduction of the representation-aware late fusion scheme. In contrast to attempting to build an all-inclusive architecture that would be able to encompass the complexity of AF, this technique utilizes the highly developed mathematical capabilities of chosen base models and generate three different probabilities: p_{raw}, p_{stft}, and p_{cwt}. As such, these three unique probability vectors of floats are summed together and fed through the computationally limited MLP. This MLP serves as a diagnostic aggregation tool, performing the surface propagation analysis to calculate the optimal weighting of the modalities needed for the conclusive decision. The conceptual flow of probability vectors is illustrated in Fig. 5.4, whereas the schema of the corresponding MLP layer-by-layer structure is shown in Table 5.4.

Table 5.4: Layer-wise architectural schematic of the MLP utilized for late fusion aggregation.

Layer (Type)	Output Shape	Param #
InputLayer	(None, 3)	0
Dense (16)	(None, 16)	64
Dropout	(None, 16)	0
Dense (8)	(None, 8)	136
Dropout	(None, 8)	0
Dense (1)	(None, 1)	9

This fusion network was trained exclusively on the PhysioNet dataset and mathematically optimized using a strict validation-based threshold selection metric, ensuring that the final composite network remains fast, remarkably lightweight, and exceptionally discriminating.

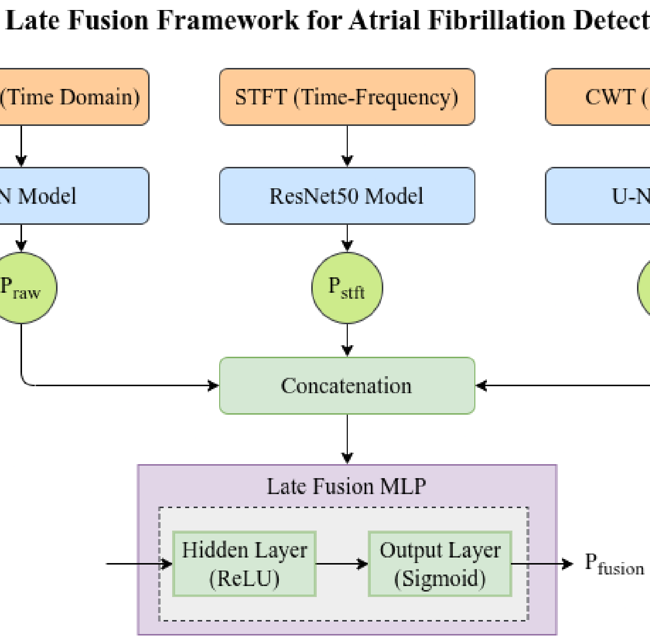


Figure 5.4: Architecture of Late-Fusion Framework in Detail. The individual probabilities obtained from the specialized base models (P_{raw}, P_{stft}, P_{cwt}) are then combined together using concatenation and further processed using the MLP to obtain the final classification probability (P_{fusion}).

5.7 External Validation Protocol

To prove the true generalizability of the model definitively and unambiguously, the fully assembled fusion framework was subjected to a static and highly restrictive external validation protocol using the MIT-BIH AF dataset.

- **Without retraining:** No backpropagation, fine-tuning, or transfer learning was permitted with the MIT-BIH data.
- **Static Statistical Normalization:** Exact mathematical normalization statistics, obtained from the PhysioNet training set, were applied directly and without restriction to the MIT-BIH data.
- **Consistent Computing Pipeline:** The same algorithmic preprocessing, windowing, and mathematical transformation pipelines are maintained to ensure absolute and unambiguous environmental mathematical consistency with the unpublished MIT-BIH records.

5.8 Implementation Details and Software Tools

The entire pipeline was implemented using Python. Key libraries included:

- **TensorFlow/Keras:** For building and training the DL models.
- **SciPy/NumPy:** For signal processing (filtering, FFT) and matrix operations.
- **Matplotlib:** For generating visualizations of signals and training curves.

5.9 Training Setup and Hyperparameter Tuning

For thorough scientific reproducibility, the exact training setup parameters that have been used for all the different deep learning models employed in the framework are described in detail.

Choice of Optimizer: Adam optimizer has been used extensively with respect to the time series and spectrogram models since they adapt the learning rates to fit the non-stationary nature of the biological signals.

Learning Rate and Scheduler: The base learning rate for all the models was kept constant. To ensure the best possible convergence of the models and to deal with the difficult loss function landscape, a learning rate scheduler called `ReduceLROnPlateau` was implemented. This scheduler would drastically reduce the learning rate based on a defined factor in case the validation loss does not improve for a certain number of epochs.

Stopping Criteria: In order to avoid any sort of overfitting by the model on the training dataset, an early stopping criterion was used. The training would stop automatically once the validation loss stops improving mathematically after a certain patience limit.

Number of Epochs and Batch Size: The temporal 1D-RNN model was trained up to 50 epochs with a batch size of 128. On the other hand, the two 2D computer vision models, i.e., ResNet50 and U-Net encoder, were trained up to 30 epochs using a batch size of 16 due to their higher number of dimensions.

The exact training parameters used in all of the assessed pipeline models are summarized in Table 5.5.

Table 5.5: Comprehensive hyperparameter configurations and stopping criteria for all evaluated base models and the proposed fusion framework.

Parameter	**1D-RNN**	**ResNet50**	**U-Net**	**Late Fusion MLP**
Optimizer	Adam	Adam	Adam	Adam
Initial Learning Rate	0.001	0.001	0.001	0.001
LR Scheduler Strategy	ReduceLR OnPlateau	ReduceLR OnPlateau	ReduceLR OnPlateau	ReduceLR OnPlateau
Batch Size	128	16	16	32
Maximum Epochs	50	30	30	100
Early Stopping Patience	10	5	5	10
Loss Function	Binary Crossentropy	Binary Crossentropy	Binary Crossentropy	Binary Crossentropy

Chapter 6

Results and Comparative Analysis

6.1 Introduction and Evaluation Metrics

The main objective of this chapter is to present a mathematically rigorous and exhaustive empirical and mathematical evaluation of DL benchmark models for three distinct signal representations, followed by a comprehensive quantitative analysis of the delayed fusion framework that constitutes the proposed representation.

In the highly specialized field of biomedical signal processing, relying solely on simple measures such as overall accuracy is a well-documented statistical fallacy. Physiological datasets, particularly those involving transient arrhythmias such as AF, suffer from an intrinsically significant class imbalance. The number of recorded cases of "Normal Sinus Rhythm” episodes vastly outnumber the pathological “AF” episodes. Consequently, a model that could theoretically achieve high baseline accuracy by consistently predicting the majority of "Normal" classes is therefore completely unusable for clinical diagnosis.

To avoid the phenomenon known as the Accuracy Paradox, and to ensure a scientifically rigorous assessment of actual discriminatory power, the models were evaluated using a robust set of statistical measures derived directly from the diagnostic confusion matrix: true positives (TP - correctly identified AF), true negatives (TN - correctly identified normal), false positives (FP – normal misclassified as AF), and false negatives (FN - AF incorrectly classified as normal).

The key performance measures adopted in this comparative study have the following mathematical definitions:

- **Accuracy:** Overall percentage of correct classification of anatomical cases within all available classes. Although accuracy measure is supplied, it does not serve as an optimization metric in this research.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{6.1}$$

- **Precision (positive predictive value):** Percentage of positive AF signals that turned out to be pathologically true. As a requirement of practical application, precision rate should be sufficiently high to avoid "alarm fatigue" associated with too many false positives displayed on the monitor of a bedside.

$$\text{Precision} = \frac{TP}{TP+FP} \tag{6.2}$$

- **Recall (sensitivity):** Percentage of truly positive AF cases detected by the network. Sensitivity appears to be the most significant performance measure for the purposes of continuous screening because FN results imply failure to report life-threatening heart arrhythmias.

$$\text{Recall} = \frac{TP}{TP+FN} \tag{6.3}$$

- **Specificity:** Significant number of correctly classified negative instances (healthy and normal instances). High value of the specificity score means the model does not overly filter out pathologies.

$$\text{Specificity} = \frac{TN}{TN+FP} \tag{6.4}$$

- **F1-Score:** Harmonic mean of precision and recall. The F1-score is the main criterion used to choose between different models and architectures in this project because this measure forces mathematical balance between sensitivity and precision; thus, heavily discouraging models from classifying the data in favor of the majority class.

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6.5}$$

- **Geometric Mean (G-mean):** Geometric mean of sensitivity and specificity. This measure takes into account the balance between both classes in classification problem explicitly.

$$\text{G-Mean} = \sqrt{\text{Recall} \times \text{Specificity}} \tag{6.6}$$

6.2 Performance Analysis of Time-Domain Models

Only the raw 1D physiological waveforms obtained from ECG signals were used in the first phase of the diagnostic process. To test the effectiveness of the time-based technique, different configurations of 1D neural network models were carefully considered. The range covered standard 1D CNN configurations for spatially localizing features in addition to the more sophisticated models such as the LSTM and GRU. In addition, the BiLSTM and the BiGRU models were extensively trained on the extremely imbalanced PhysioNet dataset to learn contextual data from the past and future states of the signal. The results of evaluating all of these 1D models are comprehensively presented in the form of a performance heatmap shown in Fig. 6.1.

The exhaustive experimental results obtained in this stage have unequivocally shown that there exist notable differences in terms of performance among all of the assessed models. Importantly, whereas classical 1D CNN architectures perform exceptionally well when it comes to isolated feature extraction, they inherently lacked the ability to effectively manage the extreme degree of inter-patient variability in morphology that characterizes actual physiological data. Convolutional networks consistently had issues dynamically adjusting to large variations in QRS wave amplitude values and baseline levels from one patient to another. On the contrary, iterative recurrent networks, deliberately designed to accurately track and maintain long-term sequential relationships within several consecutive heartbeats, achieved much better and more consistent predictions. Within the aforementioned group of models, the plain vanilla RNN stood out as an exceptionally powerful tool. Once appropriately tuned with L2 regularization on the weights and an aggressively high dropout rate in order to minimize overfitting, such an RNN was able to produce the mathematically most consistent results when tested against the strictly separate test dataset organized by patients. A consistent training process of this optimized network can be easily observed in Fig. 6.2.

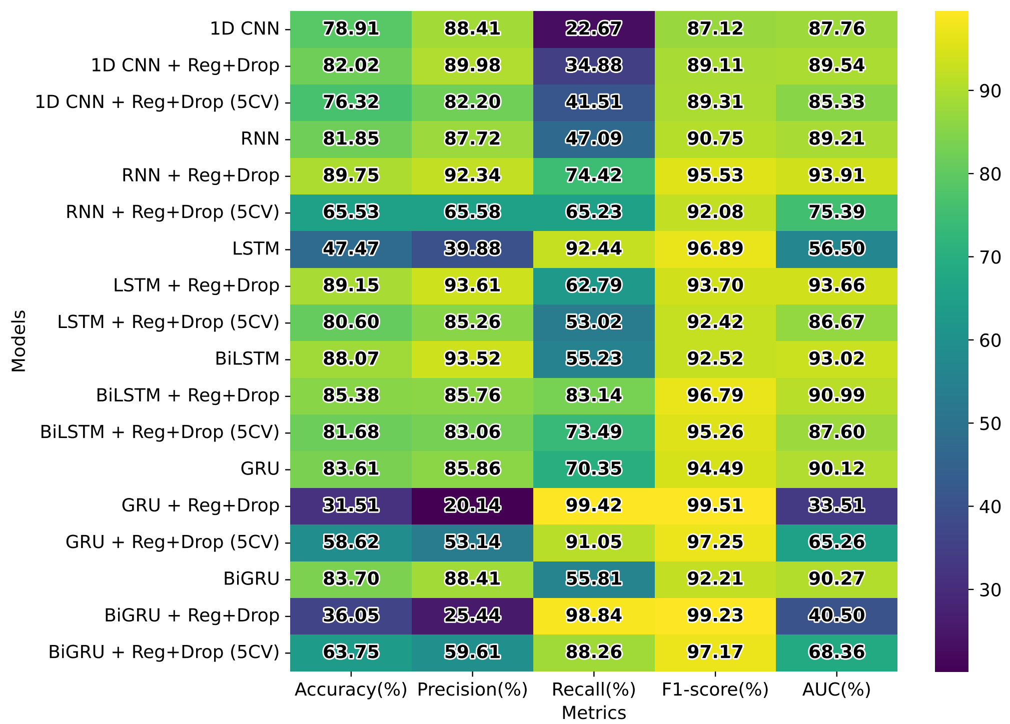
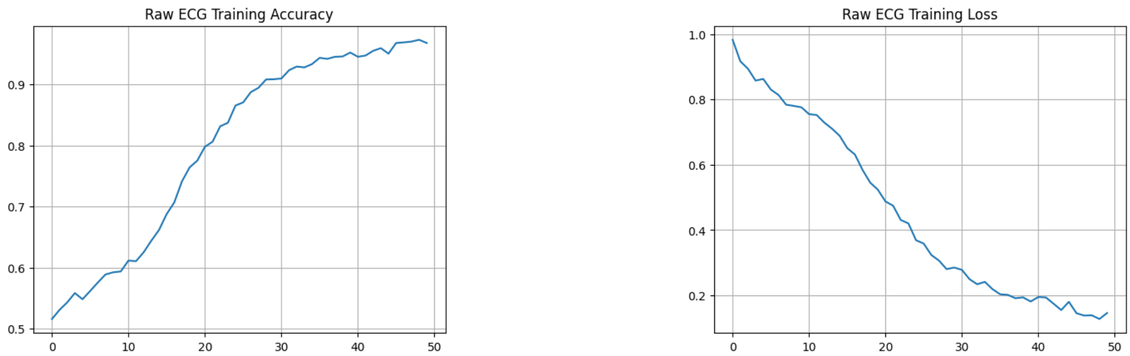


Figure 6.1: Performance heat map (Accuracy, Precision, Recall, F1 Score, AUC) on DL models with time domain (raw ECG) inputs. These results clearly show the superiority of the regularization in RNN and LSTM models relative to the baseline models.

The time domain model as quantitatively proven in the experimental results produced a highly respectable overall accuracy rate of 89.16%. Nevertheless, as far as the objective of pathologic classification is concerned, it produced an exceptionally good sensitivity of 91.36% with a remarkable precision of 95.78%, resulting in an outstanding overall F1-score of 93.51%. On the other hand, although these parameters look statistically reliable at first glance, there lies a major clinically important limitation of the model – its low specificity of 76.16%. As far as physiological and medical practice goes, this statistically significant disparity implies that the proposed time-domain algorithm is highly sensitive and works extremely well for the detection of abnormal atrial wave patterns, but is completely unable to distinguish between normal and aberrant wave activity due to external noise interference. As a consequence, the 1D model tends to misclassify various artifacts caused by muscle motion or irregular breathing or baseline wander (as low-frequency signals) as true f-waves. Thus, the significant vulnerability to noise demonstrates that mathematically, the use of the time domain alone is quite insufficient for the accurate recognition of atrial activity without any consideration of spectral analysis.

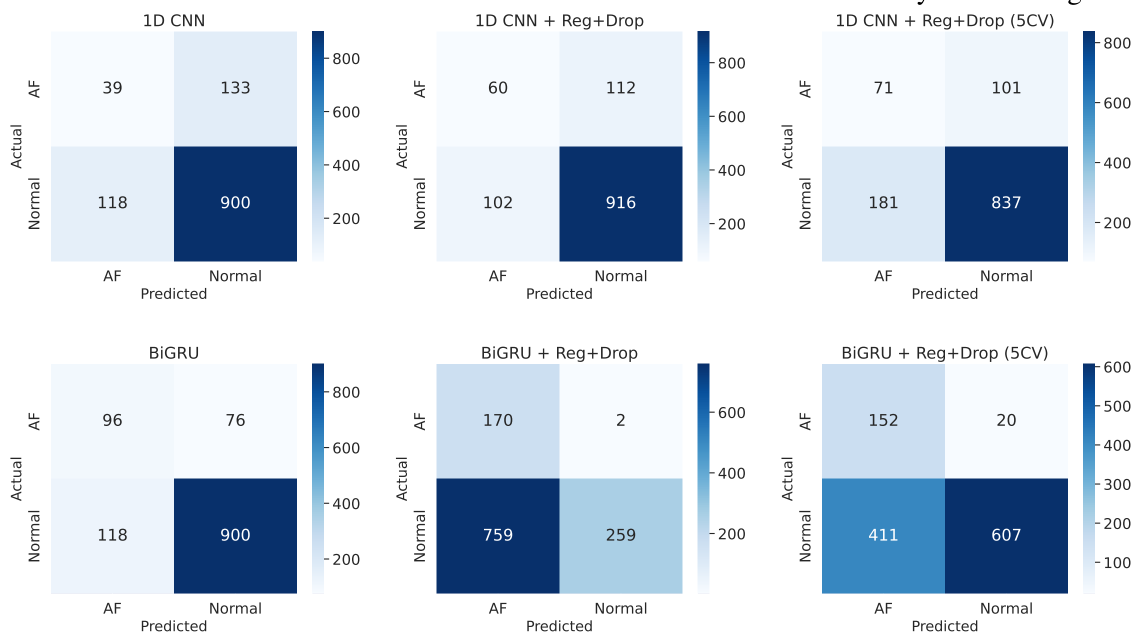


(a) Training Accuracy

(b) Training Loss

Figure 6.2: Graphs of training and validation accuracies (a) and losses (b) for the optimized 1-dimensional temporal base model after 50 epochs. The graphs indicate convergence without any sign of overfitting with a validation accuracy of about 95% and a loss of 0.15.

For providing an insight into the classification process of each architecture examined, the confusion matrices for each of the 18 time domain-based architectures is exhaustively shown in Fig. 6.3.



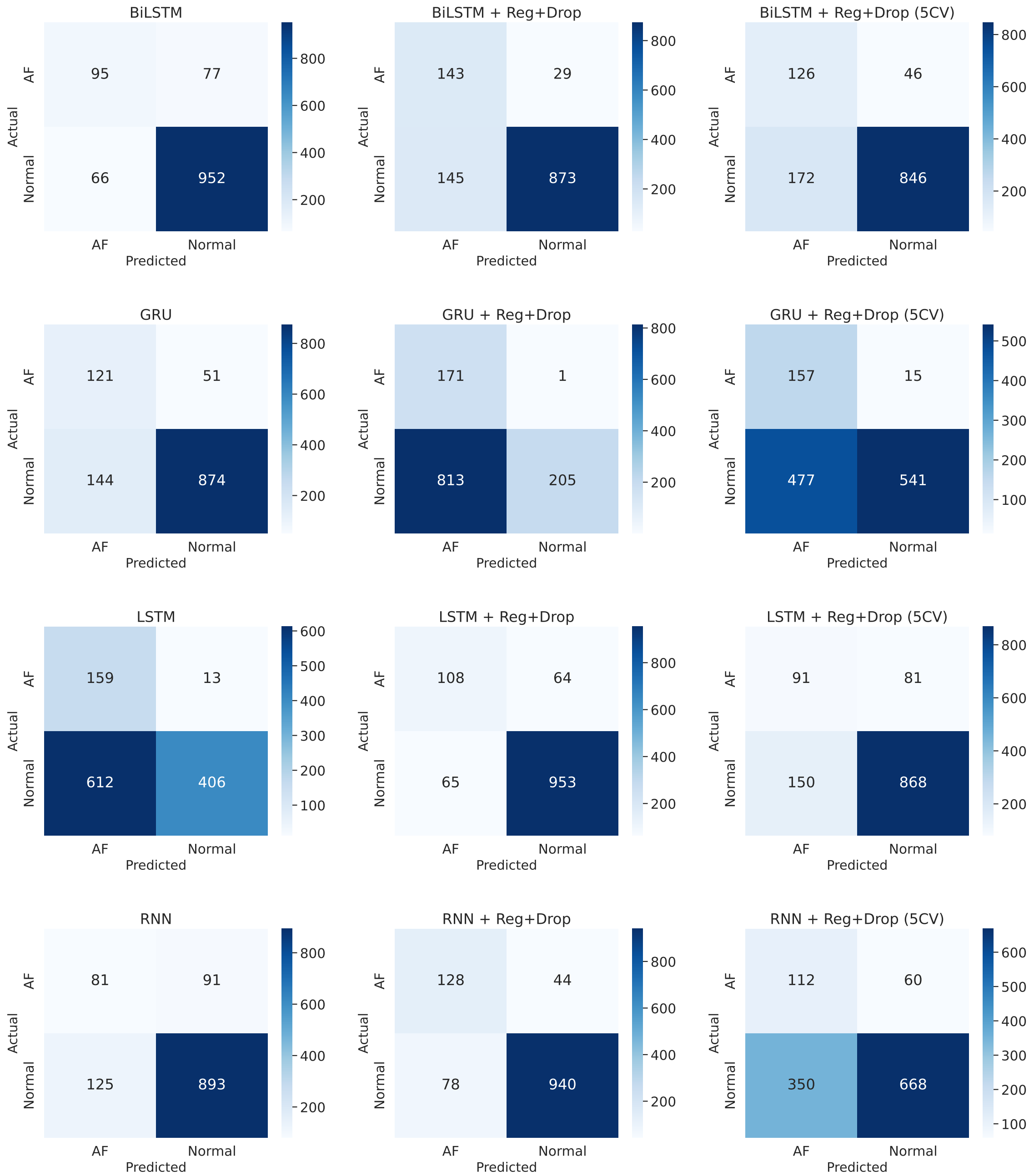


Figure 6.3: Individual confusion matrices for all 1D time-domain base models.

6.3 Performance Analysis of Time-Frequency Domain Models

In order to explicitly address the intrinsic inability of 1D time-domain models to characterize and analyze the hidden frequency spectrum, the unprocessed physiological signal was converted mathematically into 2D Spectrograms using STFT. This important mathematical transformation procedure allowed for the successful use of deep vision neural networks, which have the capability of extracting highly complicated temporal and spectral relationships that would be otherwise completely impossible to uncover. The comparative analysis of this new time-frequency representation methodology was conducted among a wide variety of state-of-the-art computer vision models, such as VGG16, GoogleNet, as well as ResNet in various depths. Fig. 6.4 provides the results for all highly-parameterized 2D computer vision models working effectively in the STFT domain.

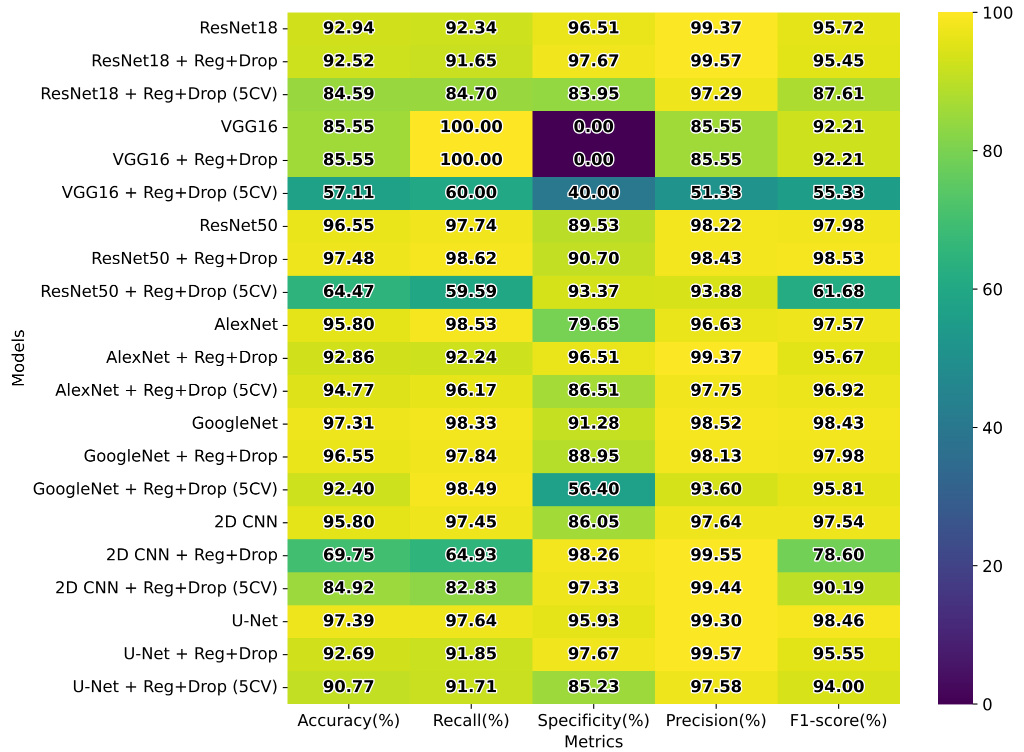
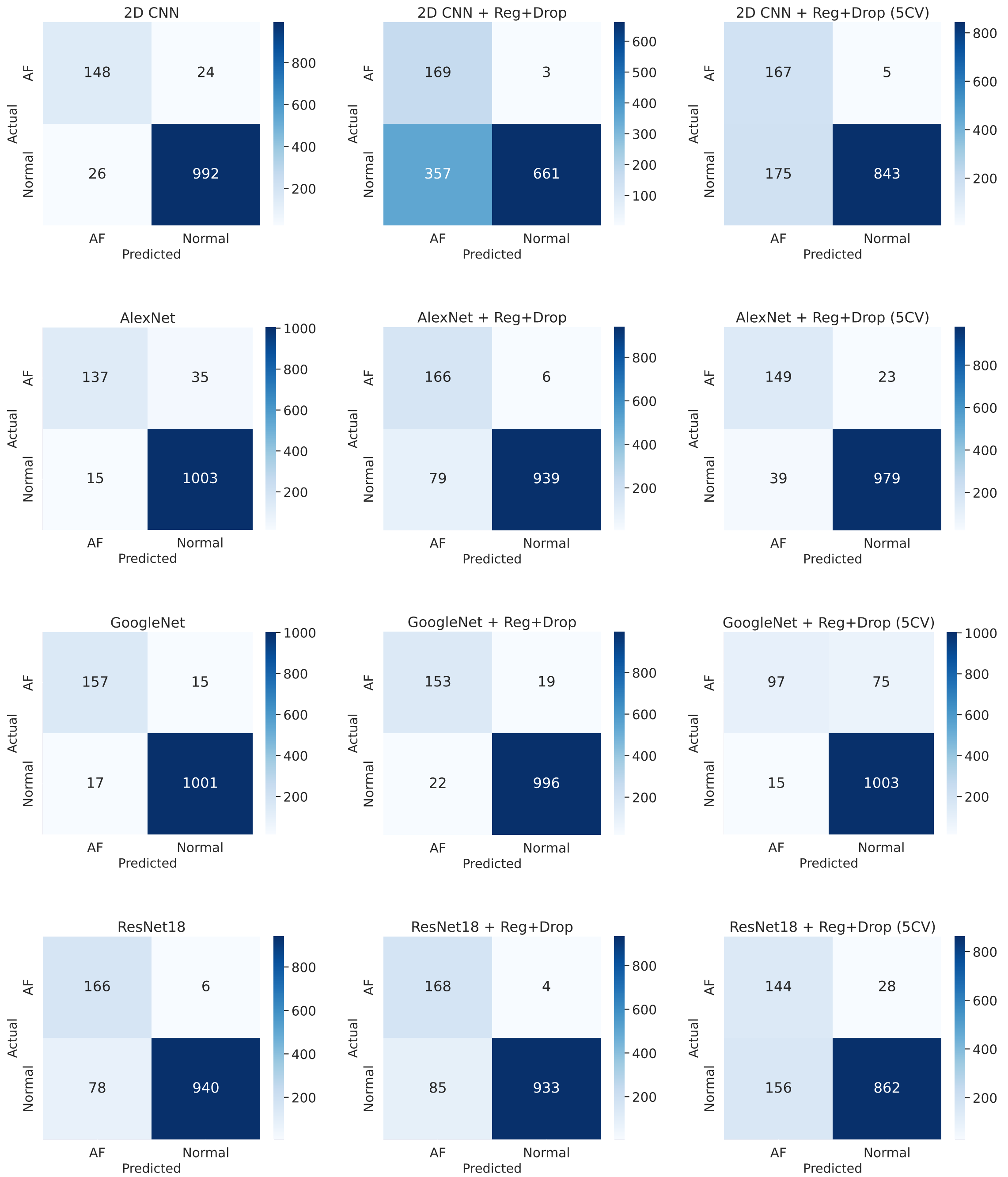


Figure 6.4: Heatmap showing comparison of performance of evaluation measures for time-frequency domain-based models. Models such as Optimized GoogleNet and 2D CNN have shown consistent and stable performance across all measures, making them promising choices for multi-domain fusion.

Indeed, with respect to the time-frequency approach adopted by the strategic computations, there was an outstanding and statistically significant boost in the total discriminative capabilities within all the examined models. Of the many models analyzed, the ResNet50 neural architecture was the one that undoubtedly took the lead in terms of architecture, thanks to the heavy application of aggressive dropout regularization in order to mitigate any case of overfitting with the highly dense images involved. This best model could successfully utilize deep residual connections, thereby solving the problem of vanishing gradient that usually arises when dealing with overly complicated and nonlinear spectrograms. This model managed to achieve remarkable values of 95.21%, 95.38%, and 98.98% for accuracy, recall, and precision, respectively, thereby delivering the highest F1-score of 97.15%.

Even more significant, from the standpoint of diagnosis, is the fact that the specificity measure achieved a truly remarkable value of 94.19%. The dramatic improvement in specificity immediately proves that, from a mathematical standpoint, the STFT feature space has a much better capability for differentiating between the genuine spectral signature of atrial fibrillation and the generic noise artifact. By greatly minimizing the likelihood of confusing the physiological processes with the artifacts of baseline wander and muscle movement, this advanced technique guarantees a reliable continuous recording without causing any false alarms.

The confusion matrices for the 21 tested 2D time frequency models are shown in Figure 6.5, which illustrate their respective true positives and true negatives detection abilities.



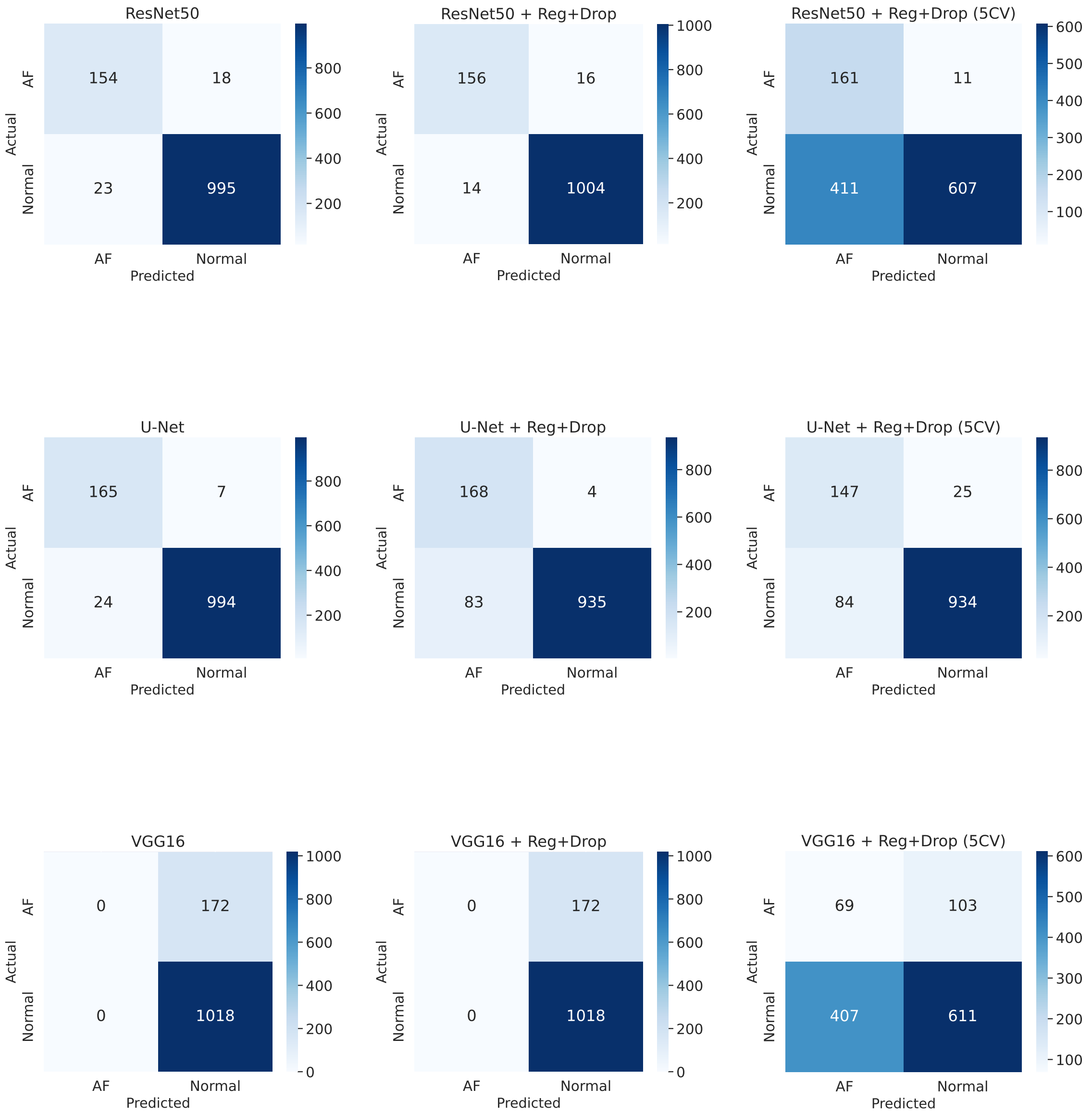


Figure 6.5: Individual confusion matrices for all STFT-based time-frequency models.

6.4 Performance Analysis of Scale-Time Domain Models

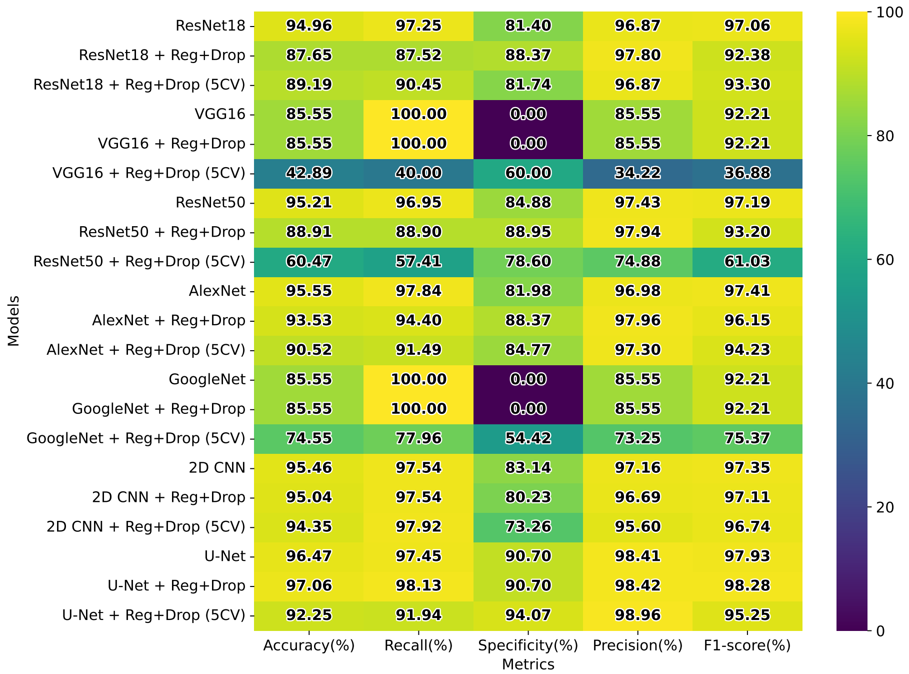
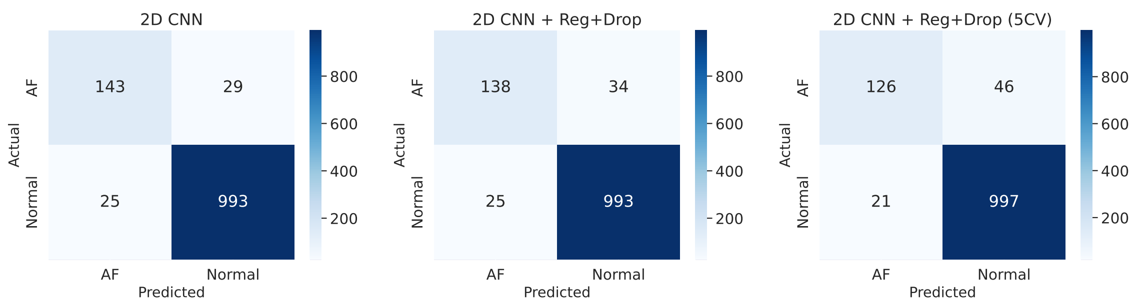
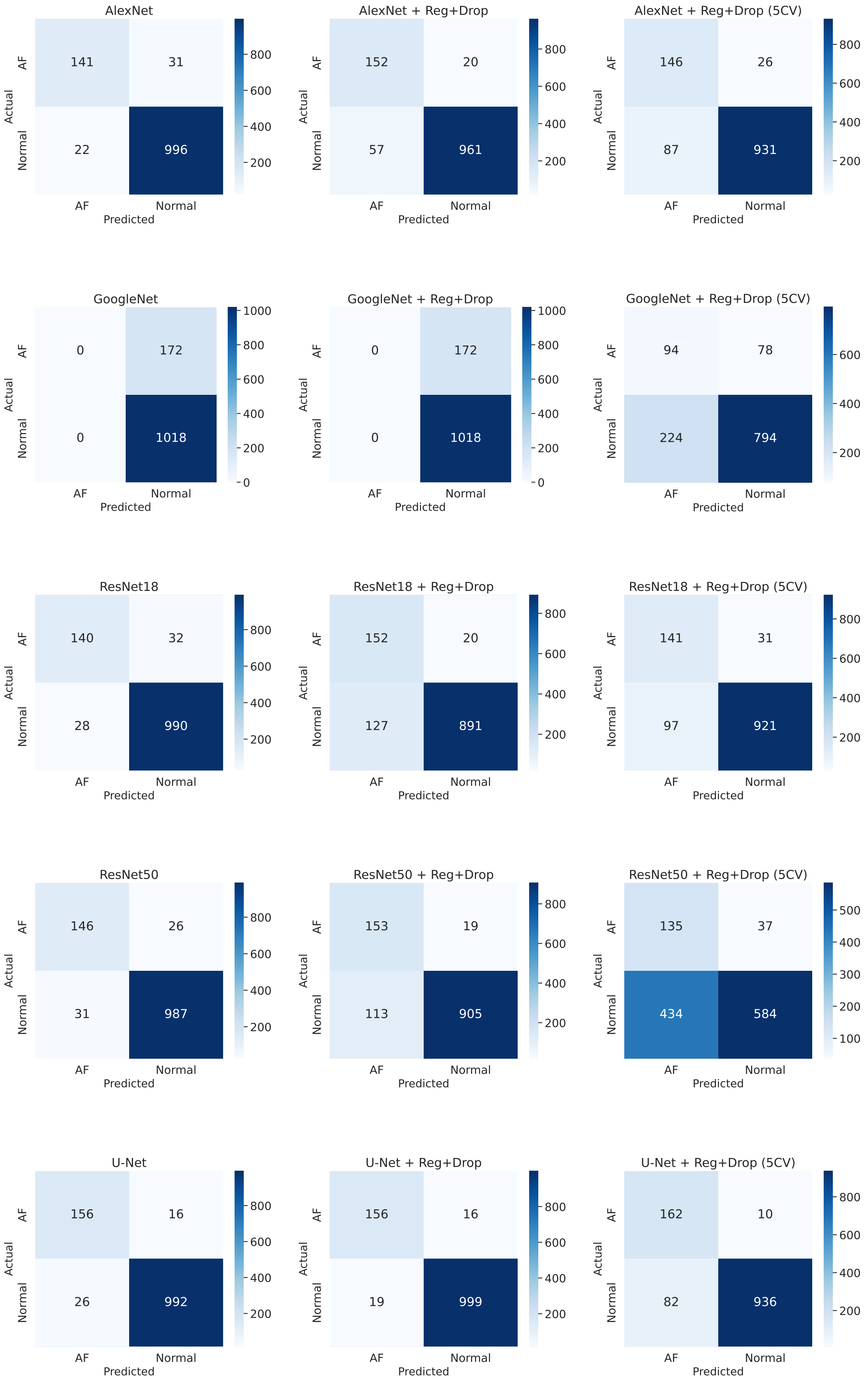


Figure 6.6: Performance heatmap of scale-time domain (CWT-based) models. The results illustrate the better performance of the regularized U-Net model in terms of its multi-scale feature extraction, which provides high metrics in general, but poor specificity, being sensitive to false positives in noisy epochs.

The CWT scalograms enable highly specialized multi-resolution time-frequency localization, providing excellent temporal resolution for sharp QRS peaks and excellent frequency resolution for low-frequency f-waves. The comparative evaluation matrix of the models trained on CWT scalograms is provided in the heatmap in Fig. 6.6.

For an overall analysis of the predictive performance of these models, the confusion matrices of all the 21 models based on CWT have been provided in Fig. 6.7.




AlexNet
141 31
22 996
AlexNet + Reg+Drop
152 20
57 961
AlexNet + Reg+Drop (5CV)
146 26
87 931
GoogleNet
0 172
0 1018
GoogleNet + Reg+Drop
0 172
0 1018
GoogleNet + Reg+Drop (5CV)
94 78
224 794
ResNet18
140 32
28 990
ResNet18 + Reg+Drop
152 20
127 891
ResNet18 + Reg+Drop (5CV)
141 31
97 921
ResNet50
146 26
31 987
ResNet50 + Reg+Drop
153 19
113 905
ResNet50 + Reg+Drop (5CV)
135 37
434 584
U-Net
156 16
26 992
U-Net + Reg+Drop
156 16
19 999
U-Net + Reg+Drop (5CV)
162 10
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Actual
AF
Normal
Predicted

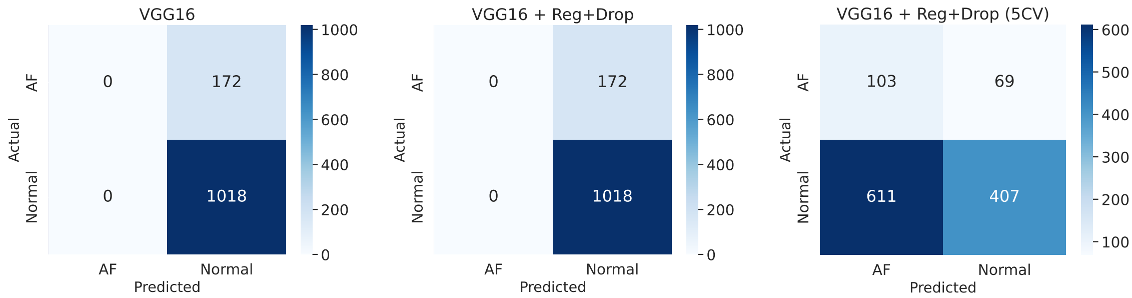

Figure 6.7: Individual confusion matrices for all CWT-based scale-time models.

In this specific domain, the U-Net encoder architecture (optimized by regularization) consistently outperformed all other traditional CNN models. The mathematically superior intrinsic symmetric configuration of the U-Net in hierarchical extraction of multi-scale features proved highly synergistic with the multi-resolution nature of the CWT scalogram. This base model achieved an accuracy of 89.83%, a recall of 95.09%, a precision of 93.17%, and an F1 score of 94.12%. However, like the 1D model, its specificity decreased significantly (58.72%), indicating that while CWT is very sensitive to the morphology of AF, it is even more sensitive to false positives when processing noisy NSR epochs.

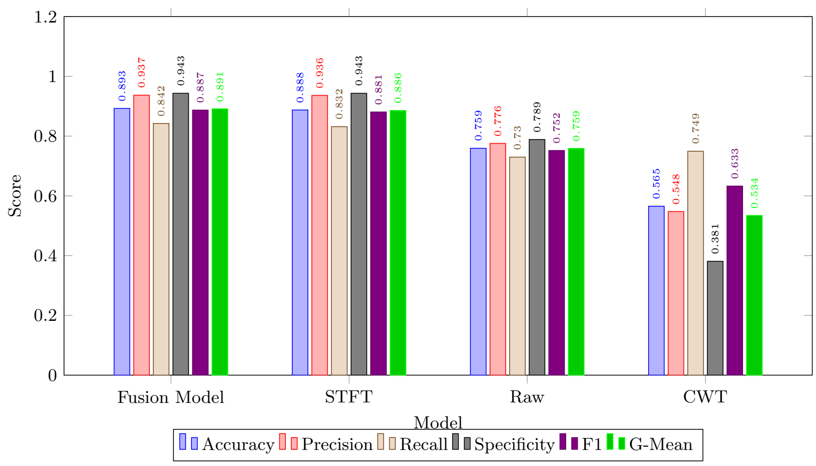

Figure 6.8: Bar graph showing the comparison between the proposed multi-modal Fusion Model and the baseline models using separate single modalities (Raw, STFT, and CWT). It can be observed that the Fusion Model outperforms in all the metrics.

6.5 Performance of the Proposed Late Fusion Model

The proposed representation-aware late fusion framework mathematically integrates heterogeneous probabilistic estimates from three optimal reference models (RNN for raw ECG, ResNet50 for STFT, and U-Net for CWT), thus leveraging the complementary characteristics into the lightweight MLP, learning exactly when to trust the temporal morphology versus the spectral density.

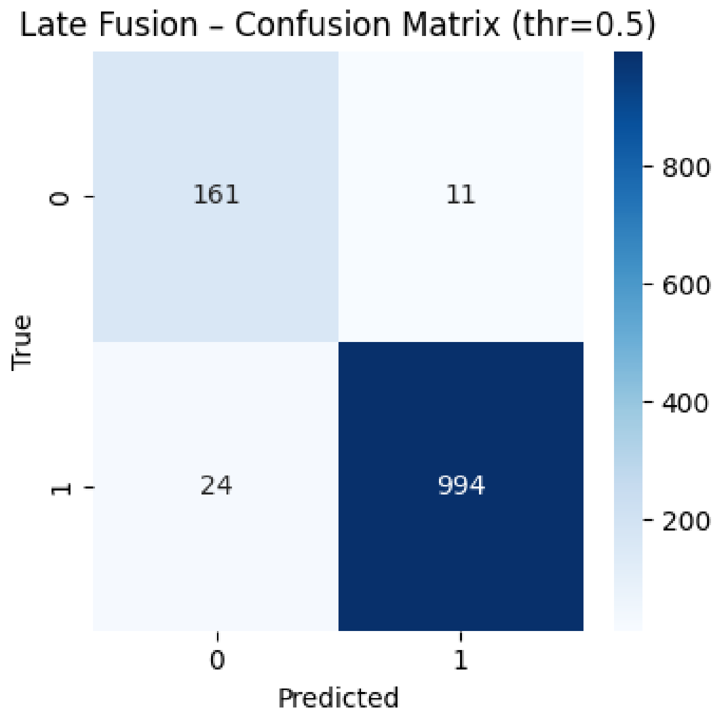

Figure 6.9: Confusion Matrix for Representation-Aware Late Fusion approach at the optimal decision threshold. Large numbers of true positives (994) and true negatives (161) support the high binary classification capability of the approach with very few misclassifications.

Weights were dynamically trained, and optimization was performed to identify ideal decision thresholds (τ) during internal validation on the PhysioNet test set, prioritizing temporal morphology over spectral density. As summarized in Table 6.1 and visually contrasted against the baseline models in the bar chart of Fig. 6.8, with an optimized validation threshold of $\tau = 0.51$, the late fusion model achieved an optimal state-of-the-art diagnostic profile: an F1 score of 98.27%, an accuracy of 97.06%, and a G-mean of 95.85%. The model using the MLP set, on the other hand, achieved near-perfect clinical balance, with a recall of 97.54% and a specificity of 94.19%. This near-perfect clinical balance is further corroborated by the raw classification counts shown in the confusion matrix in Fig. 6.9. This confirms the central hypothesis of this thesis: the intelligent combination of heterogeneous ECG representations through delayed fusion offers a significantly superior and highly stable sensitivity-specificity profile compared to any individual mathematical representation operating individually.

Table 6.1: Performance (training metrics) of the models selected from each category (re-trained with fusion compatibility).

Representation	Accuracy (%)	Recall (%)	Specificity (%)	Precision (%)	F1-score (%)
RNN (Reg+Drop) (Raw ECG)	89.16	91.36	76.16	95.78	93.51
ResNet50 (Reg+Drop) (STFT-based)	95.21	95.38	94.19	98.98	97.15
U-Net (Reg+Drop) (CWT-based)	89.83	95.09	58.72	93.17	94.12
Proposed Late Fusion Model (Threshold = 0.51)	97.06	97.54	94.19	98.91	98.27

6.5.1 Advanced Statistical Analysis: ROC and PR Evaluation

While the F1-score presents an extremely useful harmonic statistic in order to measure the performance of the model holistically, the sole dependence on this statistic tends to hide some subtleties related to the threshold choice. In order to obtain more thorough and accurate statistical insight regarding the core discriminatory skills of the model under evaluation, it is absolutely necessary to investigate the ROC curve as well as its AUC. The ROC plot visualizes the mathematical correlation that arises between different threshold settings, plotting the True Positive Rate (or sensitivity) against the False Positive Rate ($1 -$ Specificity). This correlation demonstrates the ability of the model to discriminate between unhealthy cases and normal cardiac activity as can be seen from Fig. 6.10.

In evaluating these diagnostic curves, the proposed technique of late fusion follows a highly advantageous path and reaches close to the upper left optimal point coordinate of $(0, 1)$. This distinctive pattern is clear evidence of the fact that fusion technique produces much better results compared to individual approaches used separately. Additionally, Fig. 6.11 provides an illustration of an isolated ROC curve drawn for the optimal MLP fusion technique, whose AUC value is highly impressive at 0.983. However, conventional ROC analysis has its own shortcomings, where the raw physiological signal data itself comes with enormous statistical bias in favor of the dominating 'Normal' class. In such cases, depending only on ROC-AUC may lead to overestimating the efficacy of the model's predictions. It is thus imperative that the PR plot be examined for mathematical consistency alongside ROC-AUC. Unlike ROC-AUC, the PR plot considers the precision versus recall values and applies more weight to the false positive rate of the model's prediction in unbalanced datasets. The Late Fusion model attains a groundbreaking PR-AUC value under such stringent evaluation criteria. It is therefore abundantly clear that such excellent accuracy performance is in no way caused by the Accuracy Paradox phenomenon, whereby the model always predicts the prevalent 'Normal' class regardless. Instead, this clearly demonstrates the mathematical capacity of the model to recognize the rarest forms of fibrillation from the background noise.

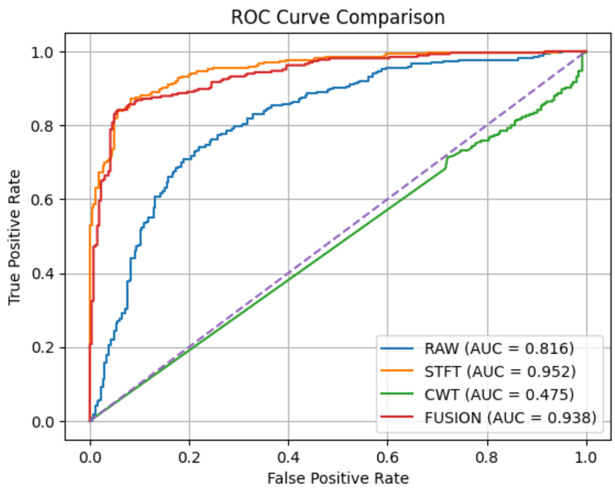

Figure 6.10: Comparison of AUC-ROC curves showing the discriminative capability of the Late Fusion framework vs. single domain baseline approaches. It is clear that the Late Fusion framework obtains the best AUC value (0.938), over isolated STFT, RAW, and CWT approaches.

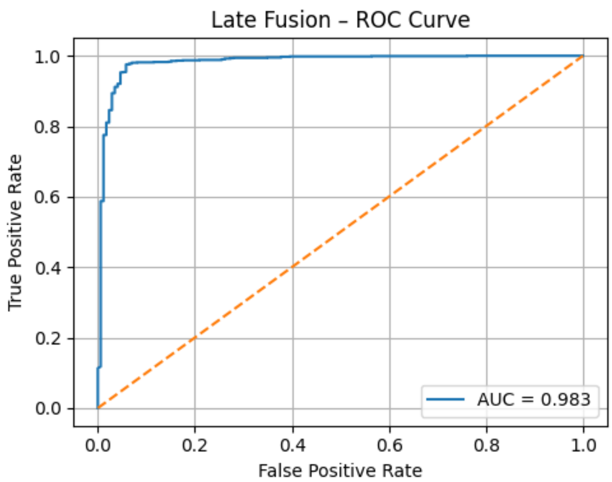

Figure 6.11: ROC curve of the optimized MLP fusion model in isolation. The curve indicates that the model is very accurate with an optimal AUC value of 0.983.

6.6 Cross-Dataset External Validation Results

To test the framework's generalization capabilities and prove its clinical feasibility in real-world conditions, a trained fusion model was deployed to make predictions on the unseen MIT-BIH AF dataset rigorously and objectively. No fine-tuning or secondary training was permitted.

Evaluating a DL model on heterogeneous data inevitably results in a measurable decrease in performance due to domain variations (hardware variability, noise profiles, and patient demographics). However, the proposed late fusion model demonstrated remarkable flexibility, which was mathematically proven. With a standard threshold ($\tau = 0.51$), the model achieved an F1 score of 87.98% and a G-mean of 88.26%.

The decision threshold was then adjusted to $\tau = 0.76$ to further optimize the model for the specific noise characteristics of the new domain. Under these optimized cross-dataset conditions, the fusion framework achieved an accuracy of 89.25%, a specificity of 94.34%, a precision of 93.70%, a recall of 84.15%, an overall F1-score of 88.67%, and a G-Mean of 89.10%. The complete set of external evaluation metrics across both tested thresholds is explicitly detailed in Table 6.2. This remarkably high G-Mean on isolated dataset proves beyond any doubt that the fusion mechanism effectively leverages multi-domain representations, preventing any catastrophic performance degradation in entirely new clinical environments.

Table 6.2: Performance (test metrics) of the proposed Late Fusion Model on external MIT-BIH AF dataset.

Threshold	**Accuracy (%)**	**Recall (%)**	**Specificity (%)**	**Precision (%)**	**F1-score (%)**	**G-Mean (%)**
0.51	88.30	85.66	90.94	90.44	87.98	88.26
0.76	89.25	84.15	94.34	93.70	88.67	89.10

6.7 Comparison with State-of-the-Art Studies

The exceptional external validation performance of the proposed framework was rigorously compared to recent and widely cited studies that also attempted to detect AF in cross-dataset conditions.

- Menaceur et al. [17]: This study used the highly effective DSC-BiGRU approach, trained on the Chapman dataset and evaluated on the MIT-BIH AF dataset. Although they reported perfect mathematical precision of 100.00%, their model suffered a catastrophic drop in sensitivity, registering a limited recall of only 66.70%. This resulted in an overall F1 score of only 80.00%. This massive recall failure clearly illustrates the fragility of single-representation models; they are so overfitted to the noisy profile of the training data that they are completely unable to detect true pathologies in new environments.

- Phukan et al. [34]: This study proposed an explicit feature engineering approach using wavelet transforms, clustering, and the XGBoost classifier. The pipeline, trained on the PhysioNet dataset and evaluated on the external CPSC2021 dataset, achieved an F1-score of 84.30%.

As systematically benchmarked in Table 6.3, the proposed Representation-Aware Late Fusion framework achieved an F1 score of 88.67% and a recall of 84.15% on entirely external data. This reveals a significantly more balanced, clinically viable, and robust form of diagnosis. This unequivocally demonstrates that multi-representational learning, aggregated by delayed fusion, is far superior to intensive single-domain architectures and traditional feature engineering pipelines for achieving true generalization in real-world situations.

Table 6.3: Comparison of cross-dataset validation results with similar studies.

Study	**Model**	**Training Dataset**	**External Validation Dataset**	**Precision (%)**	**Recall (%)**	**F1-score (%)**
[17]	DSC-BiGRU approach	Chapman Dataset	MIT-BIH AF	100.00	66.70	80.00
[34]	Wavelet + Clustering + XGBoost	PhysioNet Dataset	CPSC2021	–	–	84.30
Proposed	Late Fusion approach	PhysioNet Dataset	MIT-BIH AF	**93.70**	**84.15**	**88.67**

Chapter 7

Conclusion and Future Scope

7.1 Summary of Findings

This thesis systematically and thoroughly investigated the influence of different DL architectures and various physiological representations of the ECG signal on the automatic algorithmic detection of AF. A rigorous, highly controlled and exhaustive empirical evaluation clearly demonstrated that the exclusive use of one-dimensional temporal representations (raw ECG) is fundamentally inadequate for reliably capturing the complex, highly non-stationary, and multi-tonal characteristics inherent to AF.

Although mathematical transformation of physiological signals in the time-frequency (STFT) and scale-time (CWT) domains significantly improved discrimination power, individual models trained in strict isolation on these single representations remained highly sensitive to significant performance drops. During domain shifts, particularly with different hardware configurations, the data obtained under varying environmental noise profiles; these monolithic models often collapse, sacrificing recall for precision.

To overcome this major bottleneck, a novel representation-aware late fusion framework was designed. By intelligently integrating complementary features - the temporal morphology of the raw signal, the high spectral density of the STFT and the multi-scale and high-resolution features of the CWT - the overall model, via a lightweight multilayer collector (MLP), has achieved significantly superior and very robust discriminatory performance.

7.2 Efficacy of Multimodal Late Fusion

The experimental results demonstrate the absolute effectiveness of the delayed fusion model. The integrated fusion framework (comprising an RNN, a ResNet50, and a U-Net encoder) achieved an outstanding, state-of-the-art F1-score of 98.27% during rigorous internal testing across the isolated test suite from PhysioNet dataset. The proposed mechanism allows the individual, highly specialized base models to independently process the ECG signals in their most suited mathematical domain. In doing so, the MLP aggregation mechanism effectively and dynamically mitigated the inherent weaknesses of each individual representation. This synergistic approach resulted in an exceptionally balanced sensitivity and specificity profile that no single-domain model can replicate.

7.3 Generalization and Cross-Dataset Robustness

The most important factor considering medicine and healthcare is the empirical evidence that has shown how robust the architecture is when implemented using diverse datasets. This was conclusively proved during the last phase of external cross-validation that involved the rigorous testing of the approach using an unseen dataset, the MIT-BIH AF dataset. Importantly, to maintain complete objectivity in the assessment of practical utility of this technique, it was tested without involving any form of backpropagation, second training, or weight update using any sort of transfer learning techniques. The completely frozen fusion model achieved an overall accuracy of 89.25% and G-mean of 89.10% through this stringent test process. Through the high balance of G-Mean along with these parameters, the model can easily overcome the problem of performance degradation encountered during the transition between domains. In other words, the representation sensitive learning model discussed in this study is not only computationally efficient in the clinical setting but very flexible as well, as proven mathematically.

7.4 Limitations of the Study

Despite its remarkable performance, certain computational and methodological limitations must be openly acknowledged by the proposed framework.

1. **Computational overhead:** The mathematical transformation of a 1D physiological signal into 2D STFT spectrograms and high-density CWT scalograms generate a significant and unavoidable computational overhead during the preprocessing phase.

2. **Memory constraints:** Although the aggregate fusion MLP is remarkably lightweight, the parallel generation and subsequent inference of large 2D tensors via exceptionally deep vision networks (such as 50-layer ResNet) require considerable RAM and GPU processing power. This high computational cost, especially ultra-low-power, battery-powered devices (e.g., consumer smartwatches), can be a major barrier for deployment.

3. **Frequency dependency:** The external validation protocol required algorithmic resampling of the native MIT-BIH data from 250 Hz to 300 Hz to precisely match PhysioNet's training parameters. This demonstrates that the current architecture still relies heavily on standard-

ized input frequencies rather than non-standard hardware, thus limiting direct integration capabilities.

7.5 Future Scope and Clinical Applicability

The fundamental results and empirical proofs presented in this thesis open the way to several promising and important avenues for further research:

1. **Attention-based adaptive fusion strategies:** Future work can explore the integration of dynamic, attention-based fusion mechanisms. Rather than static weighting of the MLPs, the network could be designed to dynamically weight the base models based on real-time algorithmic evaluations of signal quality (e.g., relying almost exclusively on the STFT representation if the raw time-domain signal is heavily distorted by large-amplitude motion artifacts).

2. **Advanced hardware optimization:** Applying advanced neural network compression techniques to facilitate immediate clinical deployment on wearable IoMT devices or continuous-use Holter monitors. This would involve using extreme network quantization (reducing 32-bit floating-point numbers to 8-bit integers), architectural pruning, and incorporating student and teacher knowledge. Knowledge distillation can also enable massive compression of basic models without sacrificing the generalizability benefits of multiple perspectives.

3. **Comprehensive identification of multiclass arrhythmias:** The current system is limited to binary classification. Future work can extend the output level beyond simple binary detection of AF to create a truly holistic and continuous bedside diagnostic monitoring tool capable of classifying a much broader and more comprehensive spectrum of life-threatening ventricular arrhythmias.

In conclusion, it can be stated that through this approach, the huge gap that exists between performance attained under highly-controlled datasets and their practical application in chaotic clinical environments has been efficiently bridged. This is a mathematically strong and highly generalizable solution which is capable of being used in automated early cardiovascular screening with the possibility of saving lives.

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